

P25 J3401: Non-uniform Refinement

CryoSPARC Running Version: v4.7.1

Project Details

DETAIL	VALUE
Title	24sep20b_WT-Control-White_From-January
Description	This is a repeat collection of White Control Wildtype CFTR from the original January/February 2024 collection. Using
ID	P25
Created by user	Andy Paige

Job Details

DETAIL	VALUE
Title	New Job J3401
Description	Enter a description.
ID	J3401
Job type	Non-uniform Refinement
Status	completed
Created by user	Minou Emmad
Created	Tue Jun 09 2026 15:18:37 GMT-0400 (Eastern Daylight Time)
Queued	Tue Jun 09 2026 15:18:57 GMT-0400 (Eastern Daylight Time)
Launched	Tue Jun 09 2026 15:34:49 GMT-0400 (Eastern Daylight Time)
Started	Tue Jun 09 2026 15:34:57 GMT-0400 (Eastern Daylight Time)
Waiting	-
Killed	-
Completed	Tue Jun 09 2026 15:46:21 GMT-0400 (Eastern Daylight Time)
Failed	-
Last accessed	Wed Jun 10 2026 15:58:24 GMT-0400 (Eastern Daylight Time)
Size	3.55 GB

Inputs

particles

J3396.particles_all_classes	
BLOB	
J3396.particles_all_classes.blob.F	
CTF	
J3396.particles_all_classes.ctf.F	
ALIGNMENTS3D	

+-----+	
	J3396.particles_all_classes.alignments3D.F
+-----+	
+-----+	
	PASSTHROUGH
+-----+	
	J3396.particles_all_classes.alignments3D_multi.F
+-----+	

volume

J3396.volume_class_0	
+-----+	
	MAP
+-----+	
	J3396.volume_class_0.map.F
+-----+	

mask

No group connections

Parameters

Particle preprocessing

+-----+				
	PARAMETER		VALUE	
+-----+				
	Window dataset (real-space)		True	
+-----+				
	Window inner radius		0.85	
+-----+				
	Window outer radius		0.99	
+-----+				

Homogeneous Refinement

+-----+				
	PARAMETER		VALUE	
+-----+				
	Symmetry		C1	
+-----+				
	Symmetry relaxation method		none	
+-----+				
	Do symmetry alignment		True	
+-----+				
	Re-estimate greyscale level of input reference		True	
+-----+				
	Number of extra final passes		0	
+-----+				
	Maximum align resolution (A)		None	
+-----+				
	Initial lowpass resolution (A)		30	
+-----+				
	GSFSC split resolution (A)		20	
+-----+				
	Force re-do GS split		False	
+-----+				
	Adaptive Marginalization		True	
+-----+				
	Non-uniform refine enable		True	
+-----+				
	Non-uniform filter order		8	
+-----+				
	Non-uniform AWF		3	
+-----+				
	Enforce non-negativity		False	
+-----+				
	Window structure in real space		True	
+-----+				
	Skip interpolant premult		True	
+-----+				
	Ignore DC component		True	
+-----+				
	Ignore tilt		False	
+-----+				

Ignore trefoil	False	X		X	
Ignore tetra	False	X		X	
Use random ordering of particles	True	X		X	
Disable auto batchsize	False	X		X	
Batchsize epsilon	0.001	X		X	
Batchsize snrfactor	50	X		X	
Reset input per-particle scale	True	X		X	
Minimize over per-particle scale	False	X		X	
Scale min/use start iter	0	X		X	
Noise priorw	50	X		X	
Noise initw	200	X		X	
Noise initial sigma-scale	3	X		X	
Initialize noise model from images	False	X		X	
Dynamic mask threshold (0-1)	0.2	X		X	
Dynamic mask near (A)	6	X		X	
Dynamic mask far (A)	14	X		X	
Dynamic mask start resolution (A)	12	X			
Dynamic mask use absolute value	False	X		X	
Show plots from intermediate steps	True	X			
GPU batch size of images	None	X		X	

Defocus Refinement

PARAMETER	VALUE	DEFAULT	SPEC	ADVANCED	
Optimize per-particle defocus	False	X		X	
Num. particles to plot	3	X			
Minimum Fit Res (A)	20	X			
Defocus Search Range (A +/-)	2000	X			
GPU batch size of images	None	X		X	

Global CTF Refinement

PARAMETER	VALUE	DEFAULT	SPEC	ADVANCED	
Optimize per-group CTF params	False	X		X	
Num. groups to plot	3	X			
Binning to apply to plots	1	X		X	
Minimum Fit Res (A)	10	X			
Fit Tilt	True	X			
Fit Trefoil	True	X			

Fit Spherical Aberration	False	X			
Fit Tetrafoil	False	X			
Fit Anisotropic Mag.	False	X			
GPU batch size of images	None	X		X	

Ewald Sphere Correction

PARAMETER	VALUE	DEFAULT	SPEC	ADVANCED	
Do EWS correction	False	X		X	
EWS curvature sign	negative	X		X	

Random Seeds

PARAMETER	VALUE	DEFAULT	SPEC	ADVANCED	
Random seed	2093926102	X		X	

Compute settings

PARAMETER	VALUE	DEFAULT	SPEC	ADVANCED	
Cache particle images on SSD	True	X			
Low-Memory Mode	False	X			

Outputs

particles

ALIGNMENTS3D	
J3401.particle.alignments3D	

CTF	
J3401.particle.ctf	

BLOB (PASSTHROUGH)	
J3401.particle.blob	

ALIGNMENTS3D_MULTI (PASSTHROUGH)	
J3401.particle.alignments3D_multi	

volume

MAP	
J3401.volume.blob	

MAP_SHARP	
J3401.volume.blob	

MAP_HALF_A	

J3401.volume.blob	
MAP_HALF_B	
J3401.volume.blob	
MASK_REFINE	
J3401.volume.blob	
MASK_FSC	
J3401.volume.blob	
MASK_FSC_AUTO	
J3401.volume.blob	
PRECISION	
J3401.volume.blob	

mask

MASK_REFINE	
J3401.volume.blob	

Target

ganges.biology.columbia.edu (node)

```
> [2026-06-09 15:34:49.71 GMT-4] License is valid.
> [2026-06-09 15:34:49.71 GMT-4] Launching job on lane default target ganges.biology.columbia.edu ...
> [2026-06-09 15:34:49.72 GMT-4] Running job on master node hostname ganges.biology.columbia.edu
> [2026-06-09 15:34:57.95 GMT-4] [CPU: 78.9 MB]      [Avail: 297.62 GB] Job J3401 Started
> [2026-06-09 15:34:57.98 GMT-4] [CPU: 78.9 MB]      [Avail: 297.58 GB] Master running v4.7.1, worker running v4.7.1
> [2026-06-09 15:34:58.00 GMT-4] [CPU: 79.2 MB]      [Avail: 297.56 GB] Working in directory:
                                /mnt/nfs_HuntLabSynologyDS1621CryoEM/hCFTR/Movies/24sep20b/24
                                sep20b/rawdata/CS-24sep20b-wt-control-white-from-january/J340
                                1
> [2026-06-09 15:34:58.01 GMT-4] [CPU: 79.2 MB]      [Avail: 297.56 GB] Running on lane default
> [2026-06-09 15:34:58.01 GMT-4] [CPU: 79.2 MB]      [Avail: 297.56 GB] Resources allocated:
> [2026-06-09 15:34:58.01 GMT-4] [CPU: 79.2 MB]      [Avail: 297.56 GB]   Worker:  ganges.biology.columbia.edu
> [2026-06-09 15:34:58.01 GMT-4] [CPU: 79.2 MB]      [Avail: 297.56 GB]   CPU    :  [8, 9, 10, 11]
> [2026-06-09 15:34:58.02 GMT-4] [CPU: 79.2 MB]      [Avail: 297.56 GB]   GPU    :  [3]
> [2026-06-09 15:34:58.02 GMT-4] [CPU: 79.2 MB]      [Avail: 297.59 GB]   RAM    :  [4, 5, 6]
> [2026-06-09 15:34:58.02 GMT-4] [CPU: 79.2 MB]      [Avail: 297.61 GB]   SSD    :  True
> [2026-06-09 15:34:58.03 GMT-4] [CPU: 79.2 MB]      [Avail: 297.63 GB] -----
                                -
> [2026-06-09 15:34:58.03 GMT-4] [CPU: 79.2 MB]      [Avail: 297.67 GB] Importing job module for job type nonuniform_refine_new...
> [2026-06-09 15:35:09.62 GMT-4] [CPU: 273.5 MB]     [Avail: 299.06 GB] Job ready to run
> [2026-06-09 15:35:09.62 GMT-4] [CPU: 273.5 MB]     [Avail: 299.06 GB] *****
                                **
> [2026-06-09 15:35:09.91 GMT-4] [CPU: 287.6 MB]     [Avail: 298.92 GB] Using random seed of 2093926102
> [2026-06-09 15:35:09.92 GMT-4] [CPU: 287.6 MB]     [Avail: 298.91 GB] Loading a ParticleStack with 5294 items...
> [2026-06-09 15:35:09.92 GMT-4] [CPU: 287.6 MB]     [Avail: 298.90 GB]

                                SSD cache ACTIVE at
                                /scr/instance_ganges.biology.columbia.edu:39001 (200 GB
                                reserve) (3 TB quota)

                                Cache usage                                Amount

                                Total / Usable                            3.58 TiB / 2.73 TiB
                                Used / Free                              2.73 TiB / 666.66 MiB
                                Hits / Misses                             43.80 GiB / 0.00 B
                                Acquired / Required                       43.80 GiB / 0.00 B

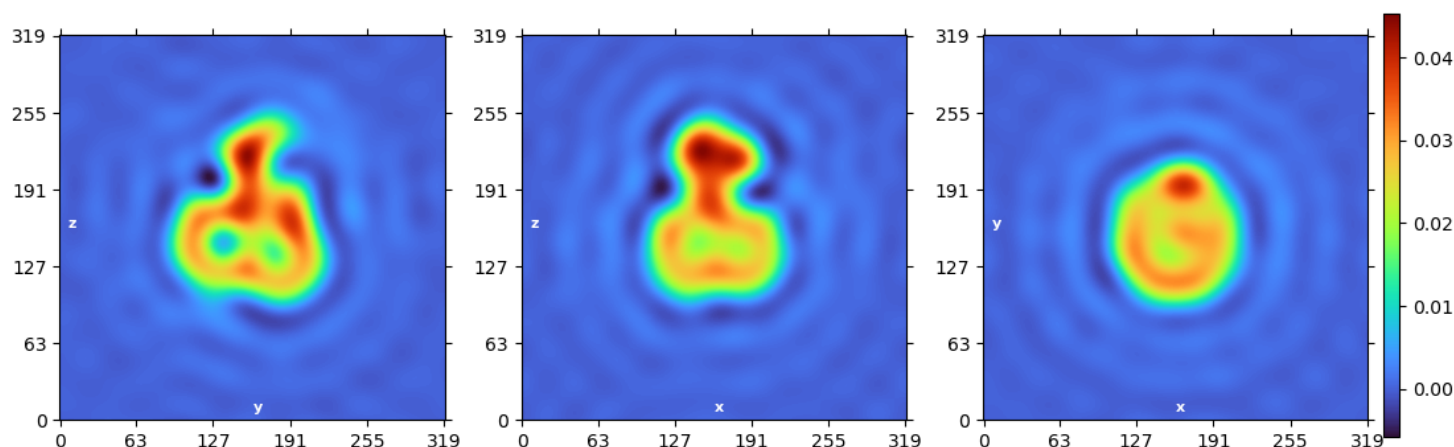
                                Progress:
                                [
                                3792/3792 (100%)
                                Elapsed: 0h 00m 04s
                                Active jobs: P25-J3401, P25-J3399, P25-J3400
                                SSD cache complete for 3792 file(s)

> [2026-06-09 15:35:14.53 GMT-4] [CPU: 331.0 MB]     [Avail: 298.13 GB] Done.
> [2026-06-09 15:35:14.53 GMT-4] [CPU: 331.0 MB]     [Avail: 298.08 GB] Windowing particles
> [2026-06-09 15:35:14.54 GMT-4] [CPU: 331.9 MB]     [Avail: 298.05 GB] Done.
> [2026-06-09 15:35:14.54 GMT-4] [CPU: 331.9 MB]     [Avail: 298.05 GB] ===== Gold Standard Split =====
> [2026-06-09 15:35:14.55 GMT-4] [CPU: 331.9 MB]     [Avail: 298.05 GB] Particles have input alignments3D connected, so reusing
                                pre-existing split
> [2026-06-09 15:35:14.55 GMT-4] [CPU: 331.9 MB]     [Avail: 298.09 GB] Set A is smaller than set B by 86 particles (1.62 percent
                                difference relative to the total dataset).
> [2026-06-09 15:35:14.58 GMT-4] [CPU: 332.4 MB]     [Avail: 298.15 GB] Split A has 2604 particles
> [2026-06-09 15:35:14.59 GMT-4] [CPU: 332.4 MB]     [Avail: 298.12 GB] Split B has 2690 particles
> [2026-06-09 15:35:14.59 GMT-4] [CPU: 332.4 MB]     [Avail: 298.08 GB] ===== Refinement =====
> [2026-06-09 15:35:14.59 GMT-4] [CPU: 332.6 MB]     [Avail: 298.10 GB] Input particles have box size 320
> [2026-06-09 15:35:14.60 GMT-4] [CPU: 332.6 MB]     [Avail: 298.11 GB] Input particles have pixel size 0.8300
> [2026-06-09 15:35:14.60 GMT-4] [CPU: 332.6 MB]     [Avail: 298.13 GB] Particles will be zeropadded/truncated to size 320 during
                                alignment
```

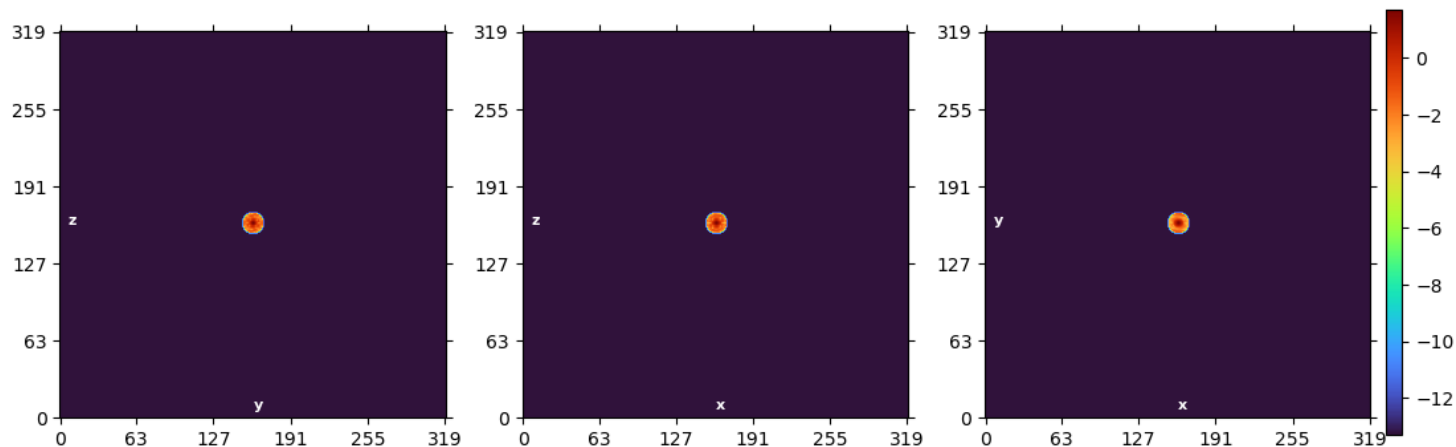
```

> [2026-06-09 15:35:14.60 GMT-4] [CPU: 332.6 MB] [Avail: 298.17 GB] Volume refinement will be done with effective box size 320
> [2026-06-09 15:35:14.60 GMT-4] [CPU: 332.6 MB] [Avail: 298.17 GB] Volume refinement will be done with pixel size 0.8300
> [2026-06-09 15:35:14.61 GMT-4] [CPU: 332.6 MB] [Avail: 298.17 GB] Particles will be zeropadded/truncated to size 320 during
backprojection
> [2026-06-09 15:35:14.61 GMT-4] [CPU: 332.6 MB] [Avail: 298.17 GB] Particles will be backprojected with box size 320
> [2026-06-09 15:35:14.61 GMT-4] [CPU: 332.6 MB] [Avail: 298.12 GB] Volume will be internally cropped and stored with box size
320
> [2026-06-09 15:35:14.62 GMT-4] [CPU: 332.6 MB] [Avail: 298.08 GB] Volume will be interpolated with box size 320 (zeropadding
factor 1.00)
> [2026-06-09 15:35:14.62 GMT-4] [CPU: 332.6 MB] [Avail: 298.05 GB] DC components of images will be ignored and volume will be
floated at each iteration.
> [2026-06-09 15:35:14.62 GMT-4] [CPU: 332.6 MB] [Avail: 298.05 GB] Spherical windowing of maps is enabled
> [2026-06-09 15:35:14.63 GMT-4] [CPU: 332.6 MB] [Avail: 298.05 GB] Refining with C1 symmetry enforced
> [2026-06-09 15:35:14.63 GMT-4] [CPU: 332.6 MB] [Avail: 298.05 GB] Refining with pose and shift marginalization during
backprojection enabled.
> [2026-06-09 15:35:14.63 GMT-4] [CPU: 332.6 MB] [Avail: 298.05 GB] Resetting input per-particle scale factors to 1.0
> [2026-06-09 15:35:14.63 GMT-4] [CPU: 332.6 MB] [Avail: 298.05 GB] Starting at initial resolution 30.000A (radwn 8.853).
> [2026-06-09 15:35:14.64 GMT-4] [CPU: 332.6 MB] [Avail: 298.05 GB] ===== Non-Uniform Refinement =====
> [2026-06-09 15:35:14.64 GMT-4] [CPU: 332.8 MB] [Avail: 298.02 GB] Non-Uniform Refinement is enabled.
> [2026-06-09 15:35:14.64 GMT-4] [CPU: 332.8 MB] [Avail: 297.97 GB] Using AWF of 3.00.
> [2026-06-09 15:35:14.65 GMT-4] [CPU: 332.8 MB] [Avail: 297.92 GB] Using butterworth filter with order 8.
> [2026-06-09 15:35:14.65 GMT-4] [CPU: 332.8 MB] [Avail: 297.91 GB] ===== Masking =====
> [2026-06-09 15:35:17.23 GMT-4] [CPU: 865.2 MB] [Avail: 295.78 GB] No mask input was connected, so dynamic masking will be
enabled.
> [2026-06-09 15:35:17.23 GMT-4] [CPU: 865.3 MB] [Avail: 295.78 GB] Dynamic mask threshold: 0.2000
> [2026-06-09 15:35:17.23 GMT-4] [CPU: 865.3 MB] [Avail: 295.77 GB] Dynamic mask near (A): 6.00
> [2026-06-09 15:35:17.24 GMT-4] [CPU: 865.3 MB] [Avail: 295.77 GB] Dynamic mask far (A): 14.00
> [2026-06-09 15:35:17.24 GMT-4] [CPU: 865.3 MB] [Avail: 295.77 GB] ===== Initial Model =====
> [2026-06-09 15:35:17.24 GMT-4] [CPU: 865.3 MB] [Avail: 295.76 GB] Resampling initial model to specified volume representation
size and pixel-size...
> [2026-06-09 15:35:20.63 GMT-4] [CPU: 1.25 GB] [Avail: 296.51 GB] Estimating scale of initial reference.
> [2026-06-09 15:35:25.59 GMT-4] [CPU: 77.4 MB] [Avail: 295.16 GB] WARNING: io_uring support disabled (not supported by kernel),
I/O performance may degrade
> [2026-06-09 15:35:26.54 GMT-4] [CPU: 1.45 GB] [Avail: 297.78 GB] Rescaling initial reference by a factor of 1.407
> [2026-06-09 15:35:28.26 GMT-4] [CPU: 1.45 GB] [Avail: 299.12 GB] Estimating scale of initial reference.
> [2026-06-09 15:35:31.39 GMT-4] [CPU: 1.42 GB] [Avail: 298.16 GB] Rescaling initial reference by a factor of 0.993
> [2026-06-09 15:35:33.08 GMT-4] [CPU: 1.42 GB] [Avail: 297.43 GB] Estimating scale of initial reference.
> [2026-06-09 15:35:35.99 GMT-4] [CPU: 1.42 GB] [Avail: 299.57 GB] Rescaling initial reference by a factor of 1.007
> [2026-06-09 15:35:38.97 GMT-4] Initial Real Space Slices

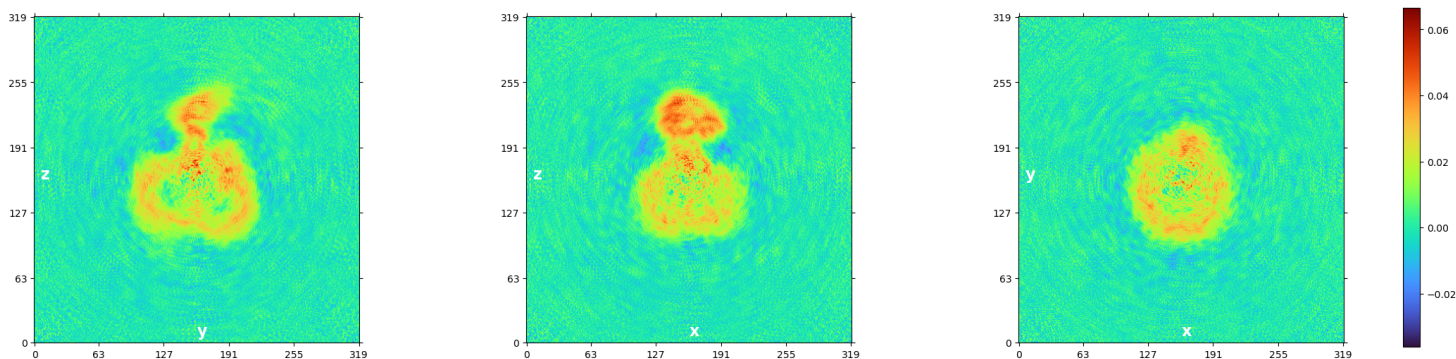
```



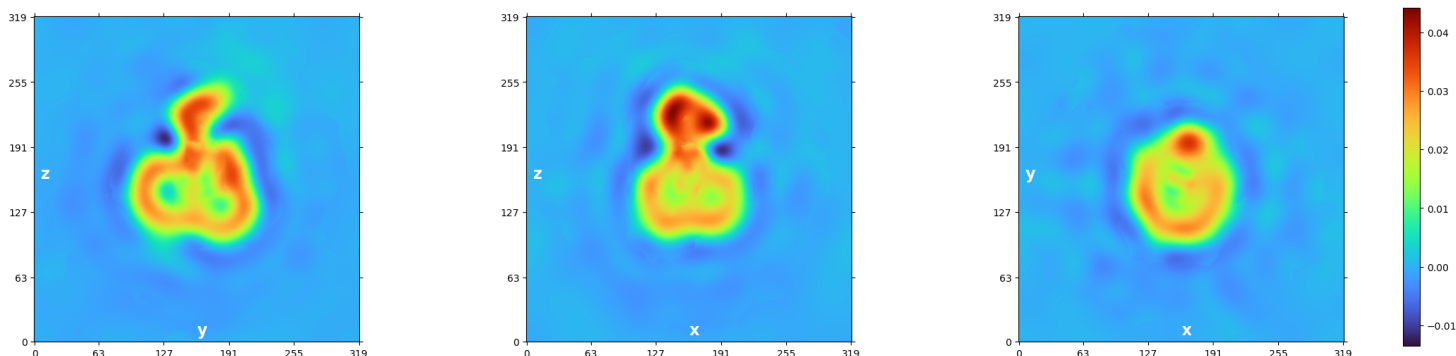
> [2026-06-09 15:35:40.76 GMT-4] Initial Fourier Space Slices



```
> [2026-06-09 15:35:40.76 GMT-4] [CPU: 1.47 GB] [Avail: 298.59 GB] ===== Starting Refinement Iterations =====
> [2026-06-09 15:35:40.77 GMT-4] [CPU: 1.47 GB] [Avail: 298.58 GB] ----- Start Iteration 0
> [2026-06-09 15:35:40.77 GMT-4] [CPU: 1.47 GB] [Avail: 298.58 GB] Using Max Alignment Radius 8.853 (30.000A)
> [2026-06-09 15:35:40.77 GMT-4] [CPU: 1.47 GB] [Avail: 298.58 GB] Auto batchsize: 2690 in each split
> [2026-06-09 15:35:44.62 GMT-4] [CPU: 1.98 GB] [Avail: 297.33 GB] -- THR 0 BATCH 500 NUM 845 TOTAL 2.9034686 ELAPSED 20.999399
--
> [2026-06-09 15:36:07.80 GMT-4] [CPU: 3.15 GB] [Avail: 295.35 GB] Processed 5294.000 images in 23.180s.
> [2026-06-09 15:36:12.14 GMT-4] [CPU: 3.48 GB] [Avail: 296.28 GB] Computing FSCs...
> [2026-06-09 15:36:12.14 GMT-4] [CPU: 3.48 GB] [Avail: 296.28 GB] Using full box size 320, downsampled box size 160, with low
memory mode disabled.
> [2026-06-09 15:36:12.14 GMT-4] [CPU: 3.48 GB] [Avail: 296.28 GB] Computing FFTs on GPU.
> [2026-06-09 15:36:14.86 GMT-4] [CPU: 3.73 GB] [Avail: 296.97 GB] Done in 2.718s
> [2026-06-09 15:36:14.86 GMT-4] [CPU: 3.73 GB] [Avail: 296.96 GB] Computing cFSCs...
> [2026-06-09 15:36:17.01 GMT-4] [CPU: 3.73 GB] [Avail: 296.93 GB] Done in 2.145s
> [2026-06-09 15:36:17.02 GMT-4] [CPU: 3.73 GB] [Avail: 296.93 GB] Using Filter Radius 26.549 (10.004A) | Previous: 8.853
(30.000A)
> [2026-06-09 15:36:23.36 GMT-4] [CPU: 4.68 GB] [Avail: 295.37 GB] Non-uniform regularization with compute option: GPU
> [2026-06-09 15:36:23.36 GMT-4] [CPU: 4.68 GB] [Avail: 295.37 GB] Running local cross validation for A ...
> [2026-06-09 15:36:52.12 GMT-4] [CPU: 7.51 GB] [Avail: 289.36 GB] Local cross validation A done in 28.756s
> [2026-06-09 15:36:53.74 GMT-4] FSC Filtered Side A
```

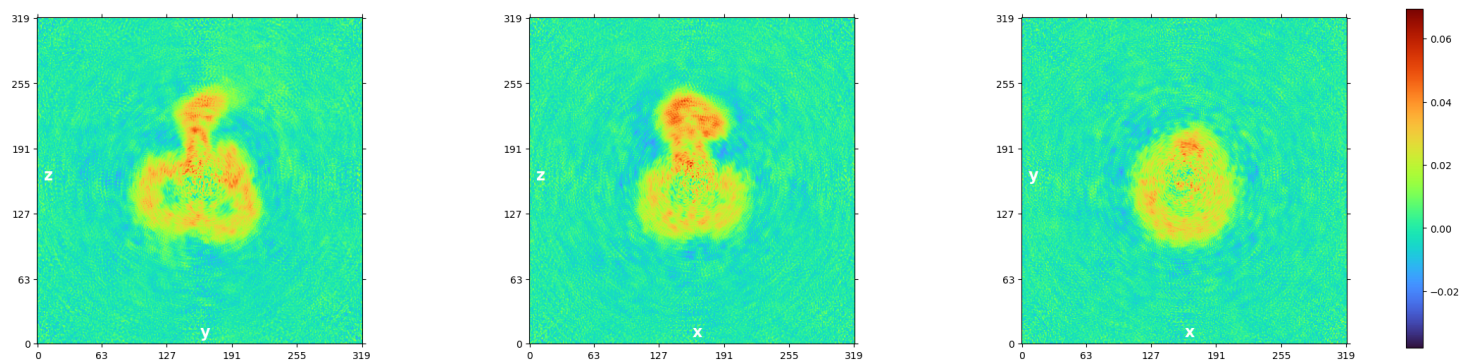


> [2026-06-09 15:36:55.06 GMT-4] CV Filtered Side A

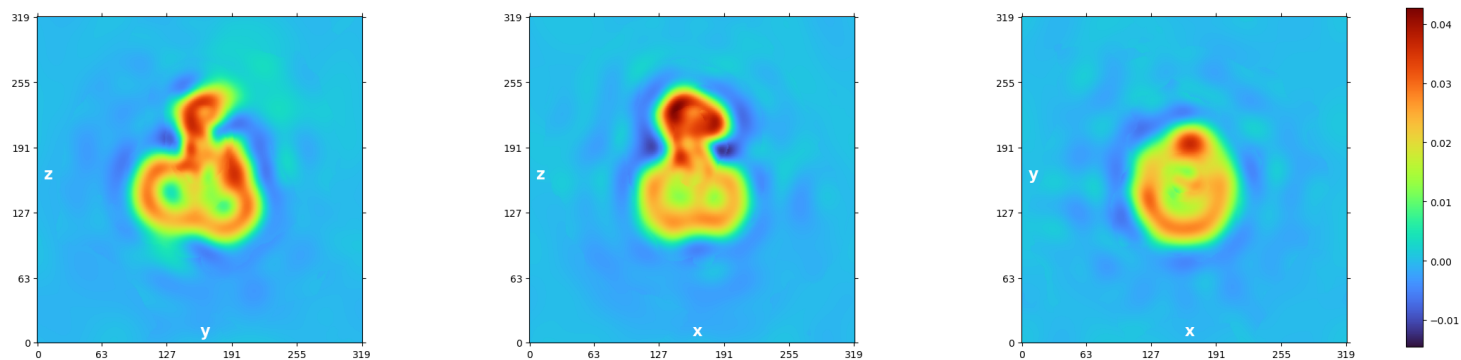


```
> [2026-06-09 15:36:55.06 GMT-4] [CPU: 7.51 GB] [Avail: 289.60 GB] Running local cross validation for B ...
> [2026-06-09 15:37:17.70 GMT-4] [CPU: 7.79 GB] [Avail: 290.77 GB] Local cross validation B done in 22.639s
```


> [2026-06-09 15:37:19.13 GMT-4] FSC Filtered Side B

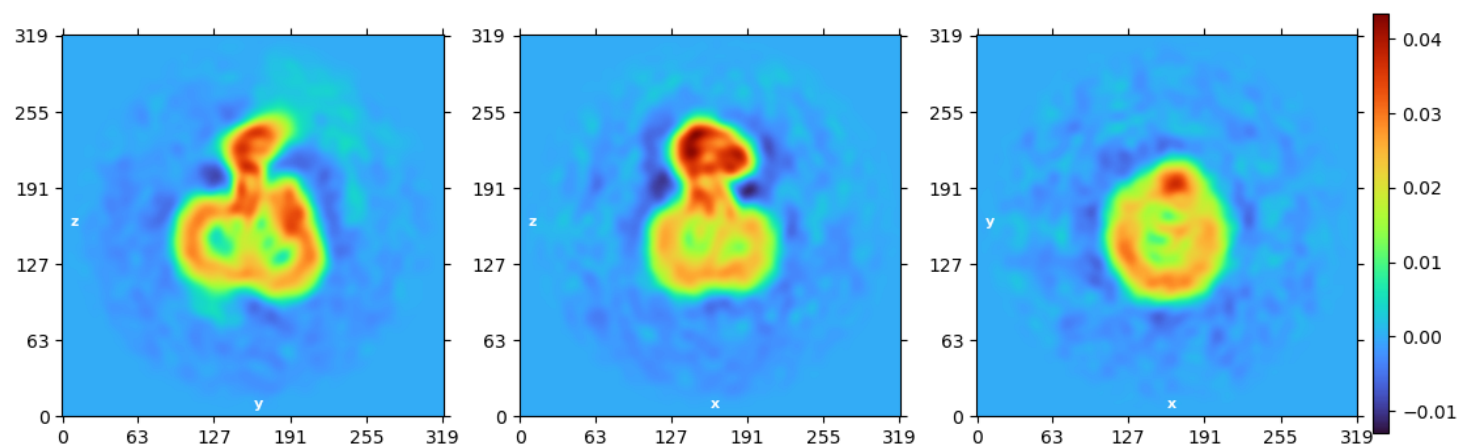


> [2026-06-09 15:37:20.27 GMT-4] CV Filtered Side B

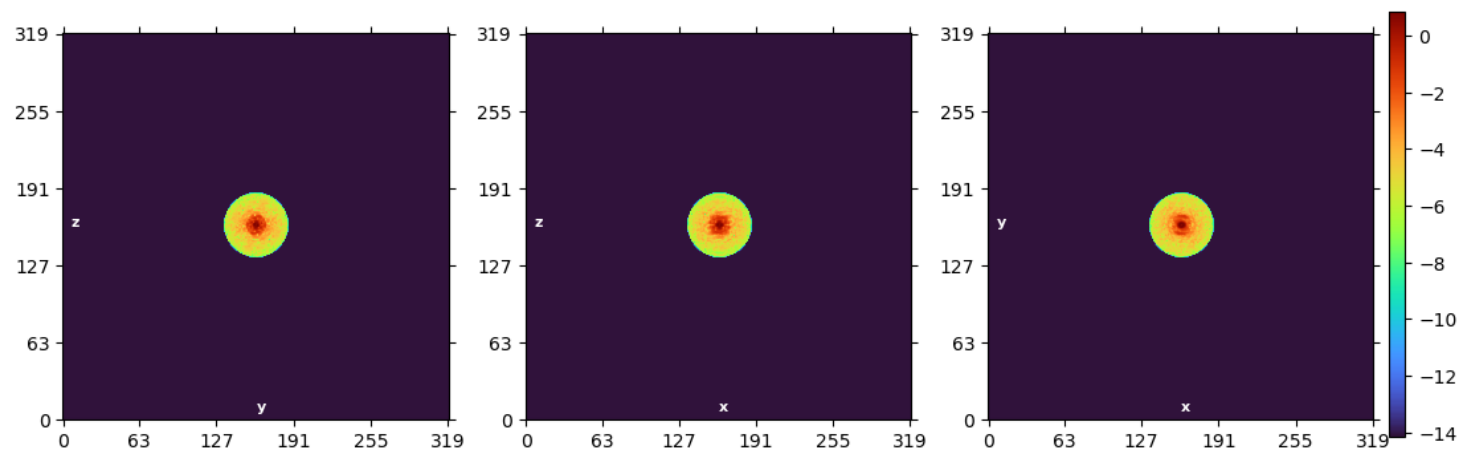


> [2026-06-09 15:37:26.23 GMT-4] [CPU: 8.02 GB] [Avail: 290.93 GB] Plotting..

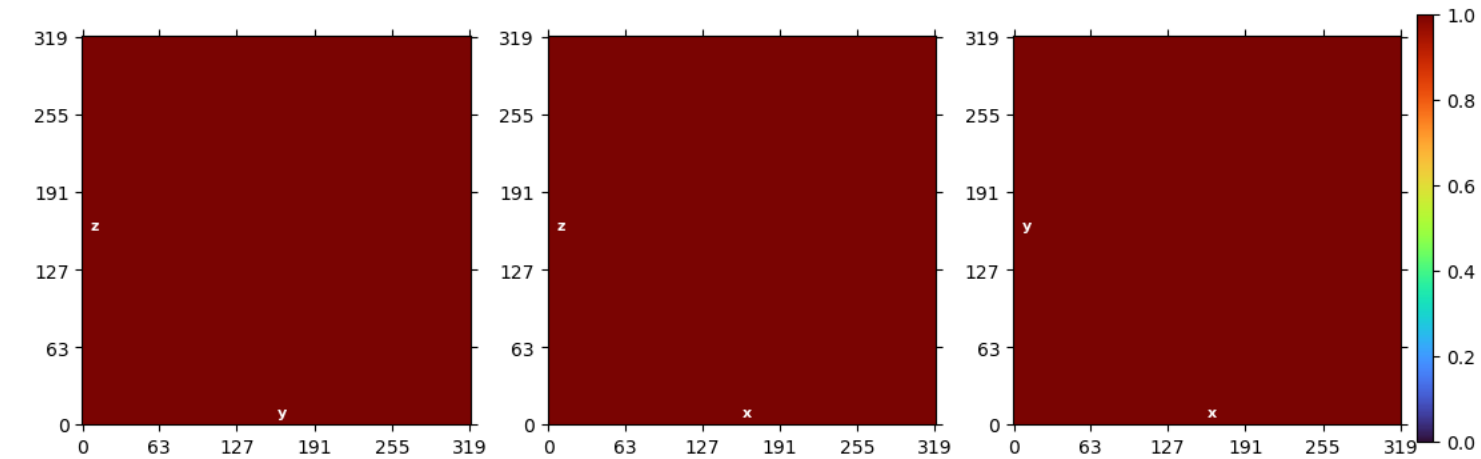
> [2026-06-09 15:37:33.18 GMT-4] Real Space Slices Iteration 000



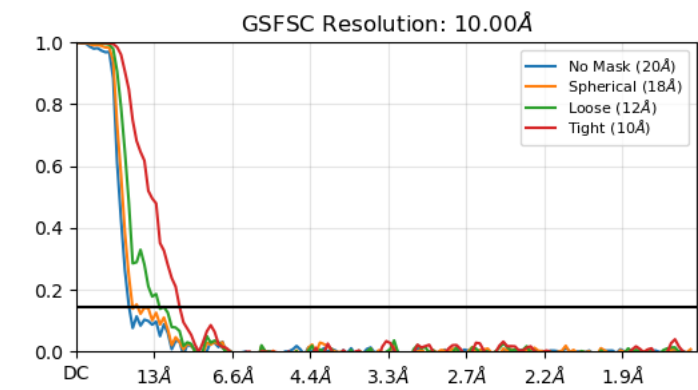
> [2026-06-09 15:37:34.65 GMT-4] Fourier Space Slices Iteration 000



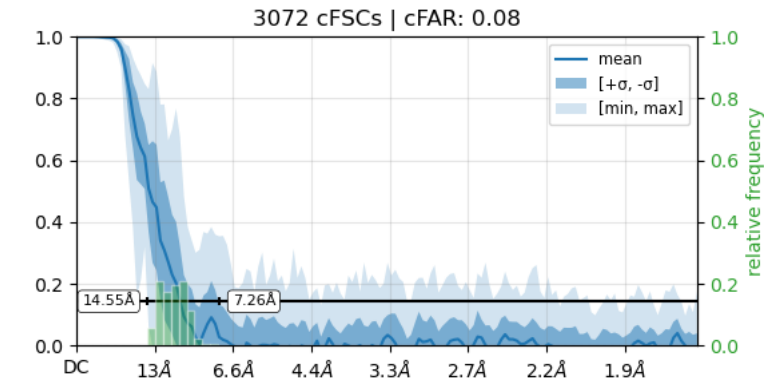
> [2026-06-09 15:37:35.73 GMT-4] Real Space Mask Slices Iteration 000



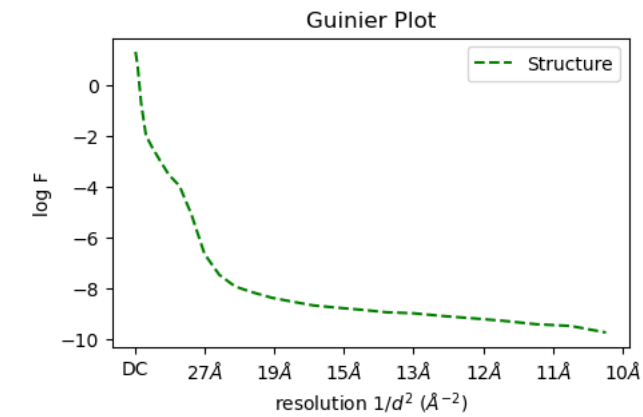
> [2026-06-09 15:37:36.48 GMT-4] FSC Iteration 000



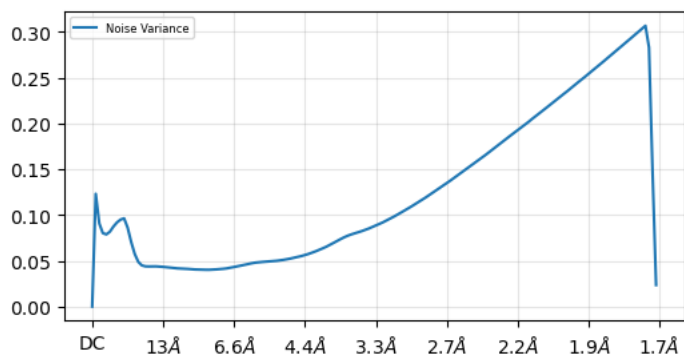
> [2026-06-09 15:37:37.77 GMT-4] cFSCs (Half-angle: 20°) Iteration 000, with tight mask



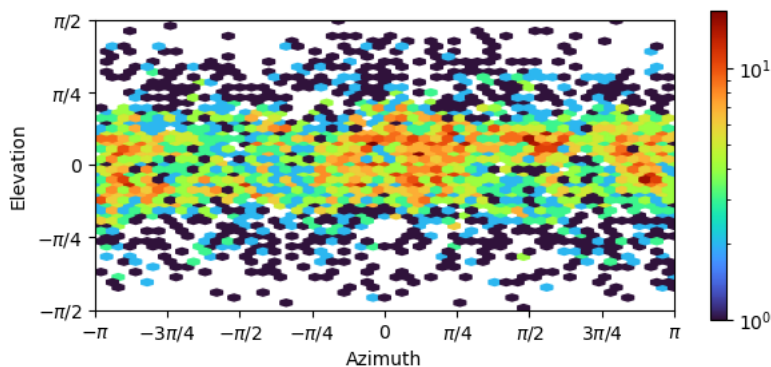
> [2026-06-09 15:37:43.02 GMT-4] Guinier Plot Iteration 000



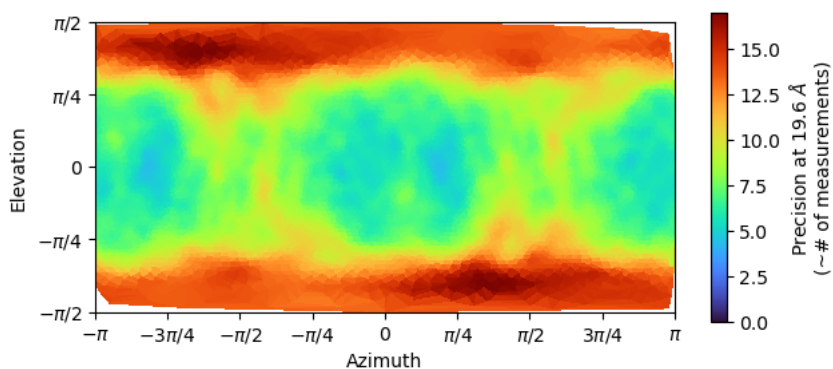
> [2026-06-09 15:37:43.39 GMT-4] Noise Model Iteration 000



> [2026-06-09 15:37:44.26 GMT-4] Viewing Direction Distribution Iteration 000

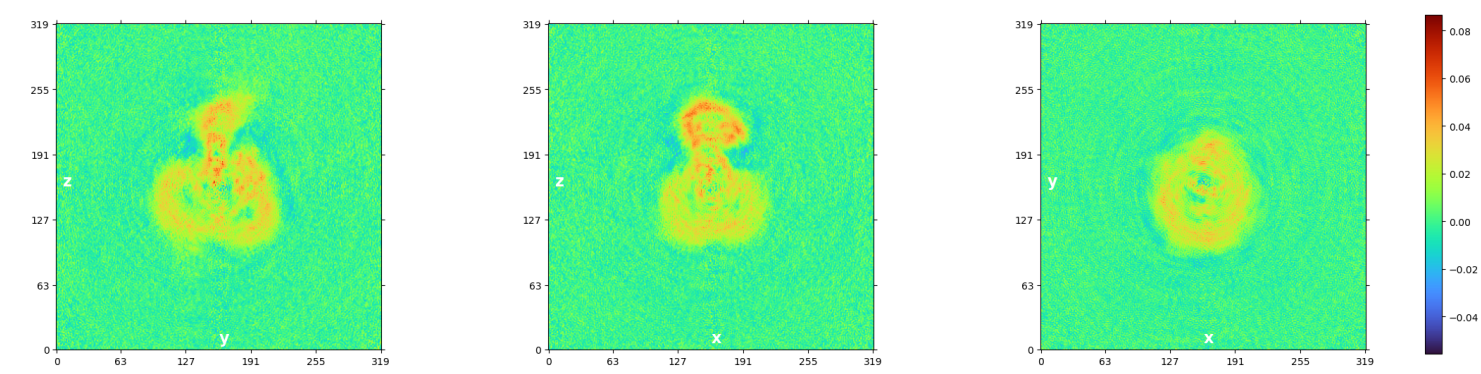


> [2026-06-09 15:37:46.04 GMT-4] Posterior Precision Directional Distribution Iteration 000

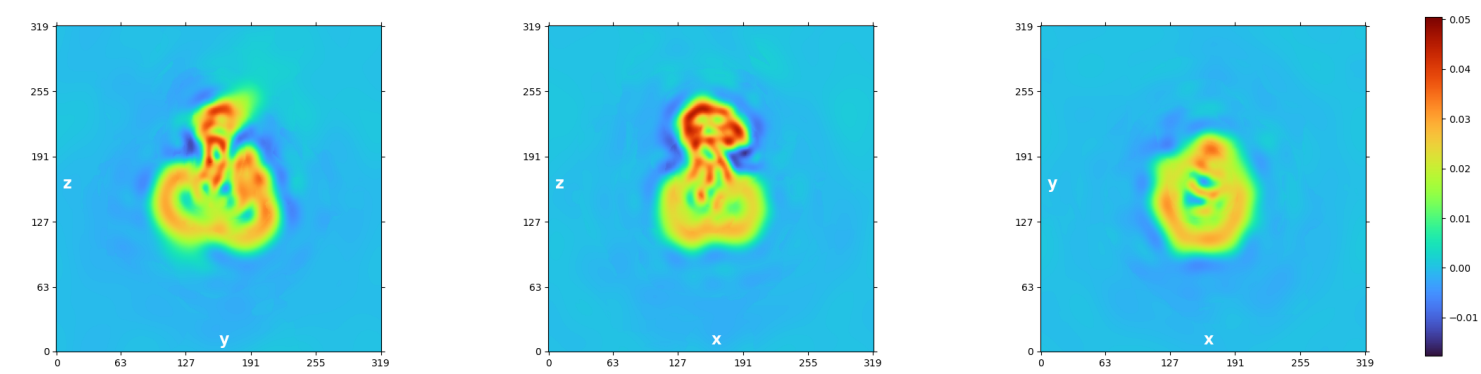


```
> [2026-06-09 15:37:46.04 GMT-4] [CPU: 8.45 GB] [Avail: 289.26 GB] Done in 19.811s.
> [2026-06-09 15:37:46.05 GMT-4] [CPU: 8.45 GB] [Avail: 289.30 GB] Outputting files..
> [2026-06-09 15:38:01.18 GMT-4] [CPU: 8.89 GB] [Avail: 286.52 GB] Done in 15.134s.
> [2026-06-09 15:38:01.19 GMT-4] [CPU: 8.89 GB] [Avail: 286.51 GB] Done iteration 0 in 140.416s. Total time so far 171.539s
> [2026-06-09 15:38:02.01 GMT-4] [CPU: 8.89 GB] [Avail: 286.82 GB] ----- Start Iteration 1
> [2026-06-09 15:38:02.01 GMT-4] [CPU: 8.89 GB] [Avail: 286.82 GB] Using Max Alignment Radius 26.549 (10.004A)
> [2026-06-09 15:38:02.01 GMT-4] [CPU: 8.89 GB] [Avail: 286.82 GB] Auto batchsize: 2690 in each split
> [2026-06-09 15:38:02.02 GMT-4] [CPU: 8.89 GB] [Avail: 286.82 GB] Using dynamic mask.
> [2026-06-09 15:38:34.20 GMT-4] [CPU: 9.39 GB] [Avail: 286.32 GB] -- THR 0 BATCH 500 NUM 845 TOTAL 3.0255224 ELAPSED 18.990451
--
> [2026-06-09 15:38:55.89 GMT-4] [CPU: 10.24 GB] [Avail: 288.39 GB] Processed 5294.000 images in 21.691s.
> [2026-06-09 15:38:59.40 GMT-4] [CPU: 10.24 GB] [Avail: 289.67 GB] Computing FSCs...
> [2026-06-09 15:38:59.40 GMT-4] [CPU: 10.24 GB] [Avail: 289.66 GB] Using full box size 320, downsampled box size 160, with low
memory mode disabled.
> [2026-06-09 15:38:59.41 GMT-4] [CPU: 10.24 GB] [Avail: 289.65 GB] Computing FFTs on GPU.
> [2026-06-09 15:39:01.41 GMT-4] [CPU: 10.24 GB] [Avail: 289.46 GB] Done in 2.013s
> [2026-06-09 15:39:01.42 GMT-4] [CPU: 10.24 GB] [Avail: 289.46 GB] Computing cFSCs...
> [2026-06-09 15:39:03.53 GMT-4] [CPU: 10.24 GB] [Avail: 289.24 GB] Done in 2.103s
> [2026-06-09 15:39:03.53 GMT-4] [CPU: 10.24 GB] [Avail: 289.24 GB] Using Filter Radius 33.580 (7.909A) | Previous: 26.549
(10.004A)
> [2026-06-09 15:39:09.82 GMT-4] [CPU: 9.49 GB] [Avail: 288.75 GB] Non-uniform regularization with compute option: GPU
> [2026-06-09 15:39:09.82 GMT-4] [CPU: 9.49 GB] [Avail: 288.79 GB] Running local cross validation for A ...
> [2026-06-09 15:39:31.56 GMT-4] [CPU: 9.74 GB] [Avail: 301.70 GB] Local cross validation A done in 21.735s
```


> [2026-06-09 15:39:32.94 GMT-4] FSC Filtered Side A



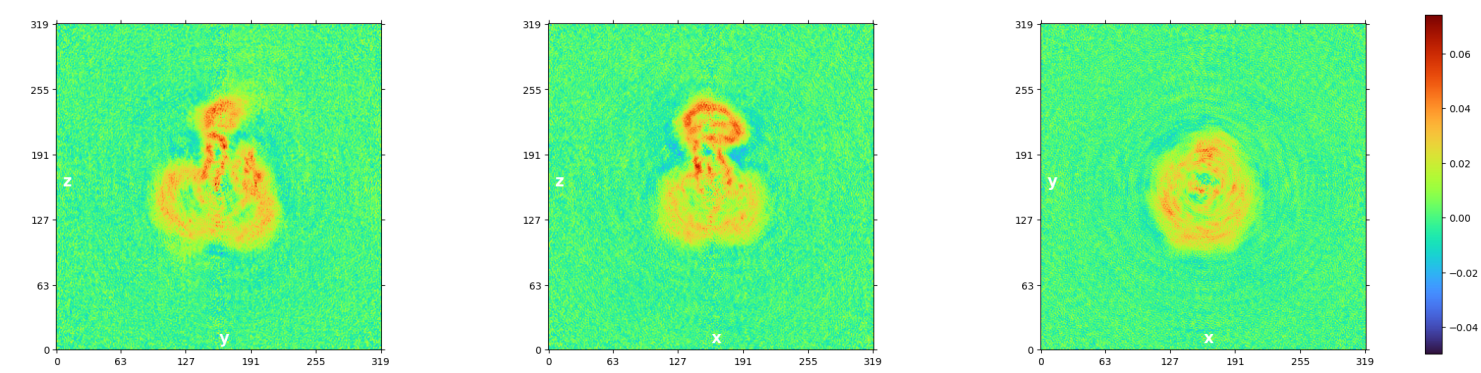
> [2026-06-09 15:39:34.08 GMT-4] CV Filtered Side A



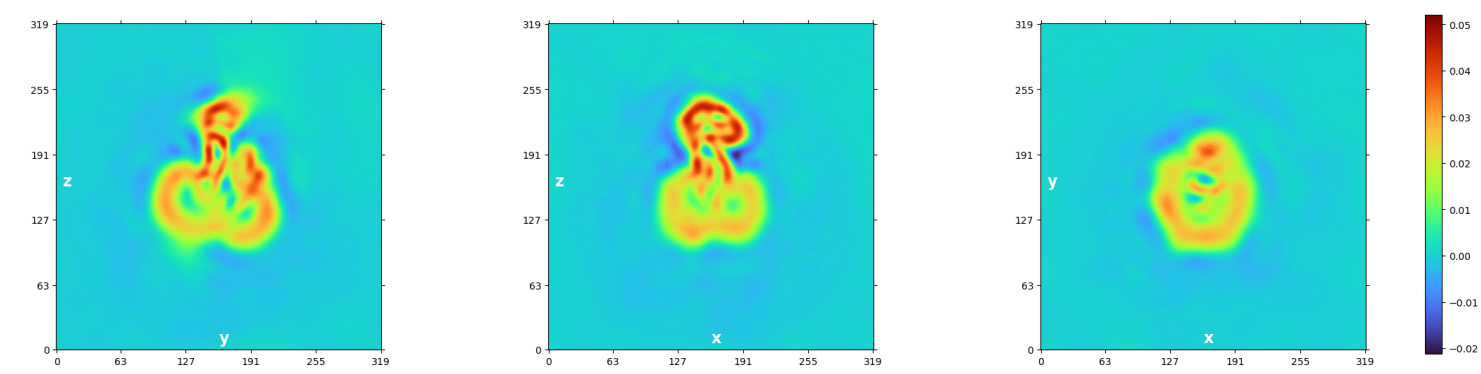
> [2026-06-09 15:39:34.08 GMT-4] [CPU: 9.74 GB] [Avail: 306.10 GB] Running local cross validation for B ...

> [2026-06-09 15:39:55.71 GMT-4] [CPU: 9.99 GB] [Avail: 313.95 GB] Local cross validation B done in 21.631s

> [2026-06-09 15:39:57.20 GMT-4] FSC Filtered Side B



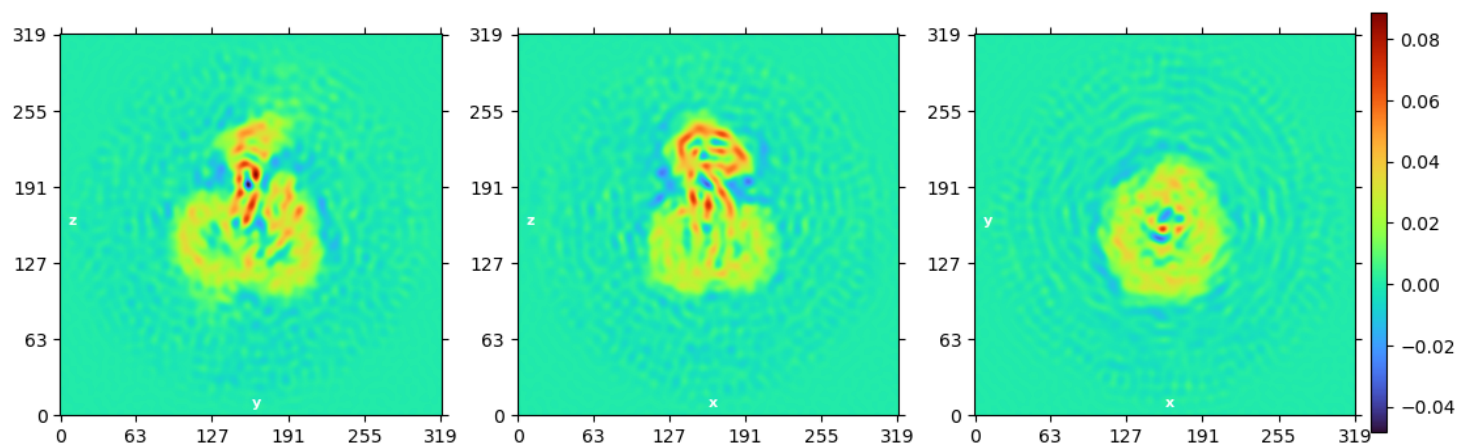
> [2026-06-09 15:39:58.30 GMT-4] CV Filtered Side B



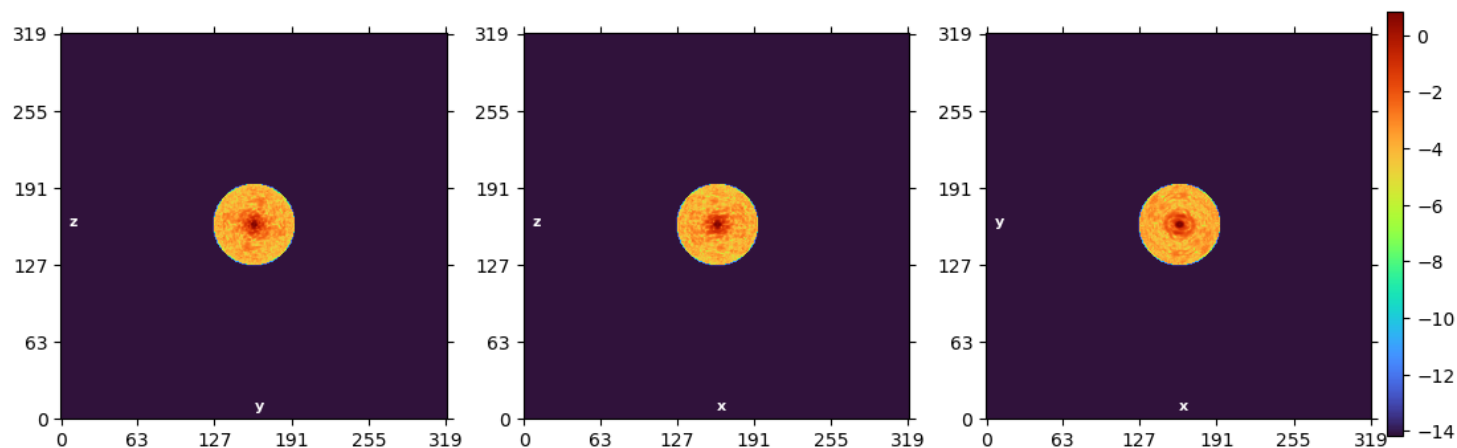
> [2026-06-09 15:40:08.16 GMT-4] [CPU: 10.21 GB] [Avail: 313.25 GB] Estimated Bfactor: -637.1

> [2026-06-09 15:40:08.17 GMT-4] [CPU: 10.21 GB] [Avail: 313.25 GB] Plotting..

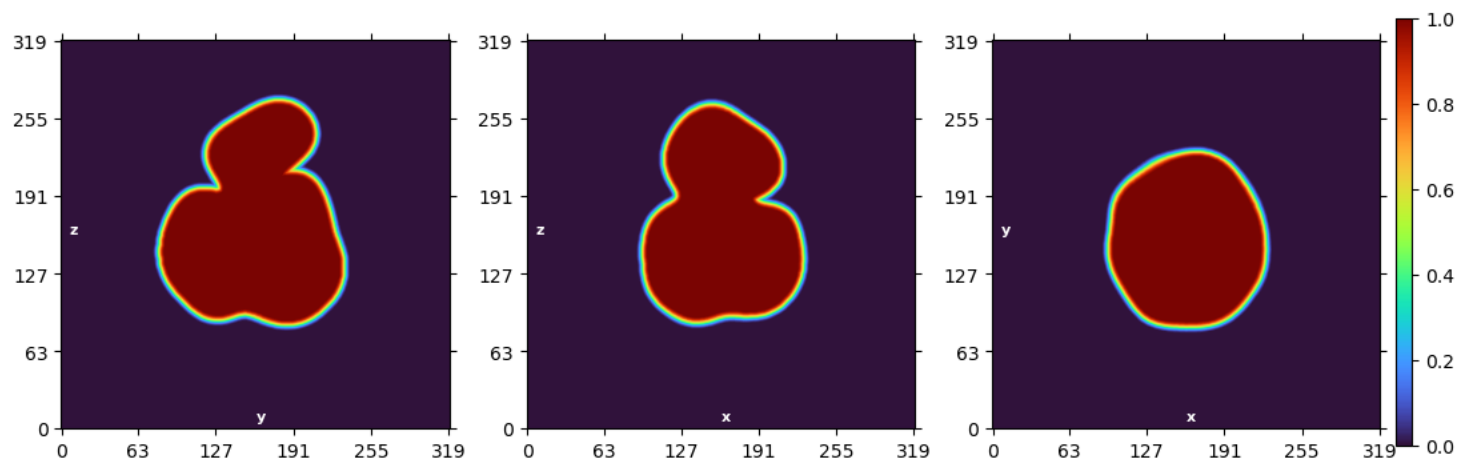
> [2026-06-09 15:40:14.36 GMT-4] Real Space Slices Iteration 001



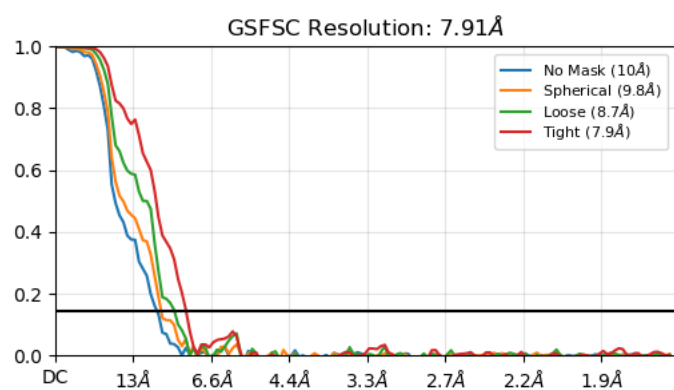
> [2026-06-09 15:40:15.62 GMT-4] Fourier Space Slices Iteration 001



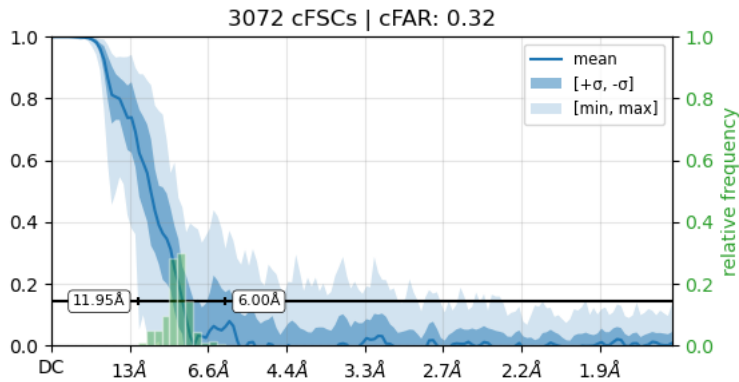
> [2026-06-09 15:40:17.47 GMT-4] Real Space Mask Slices Iteration 001



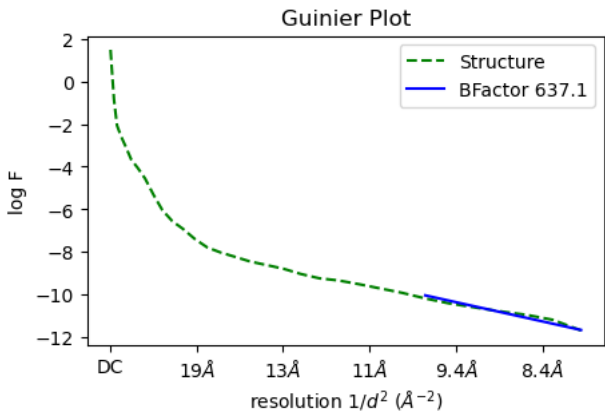
> [2026-06-09 15:40:18.01 GMT-4] FSC Iteration 001



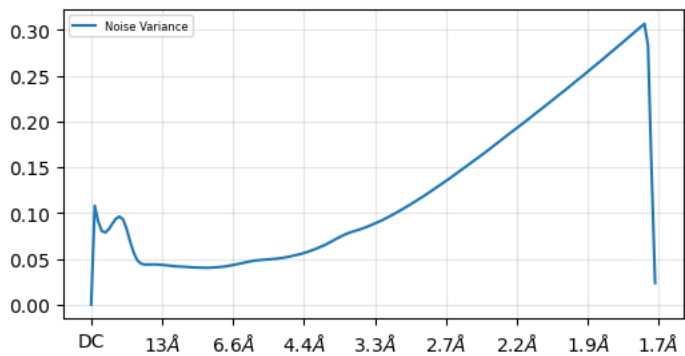
> [2026-06-09 15:40:18.78 GMT-4] cFSCs (Half-angle: 20°) Iteration 001, with tight mask



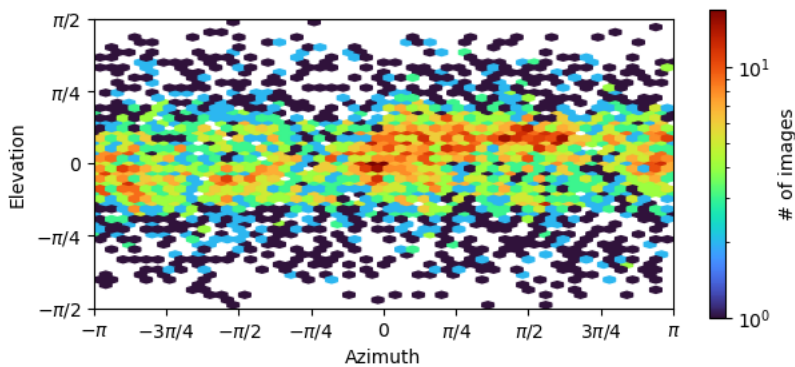
> [2026-06-09 15:40:24.42 GMT-4] Guinier Plot Iteration 001



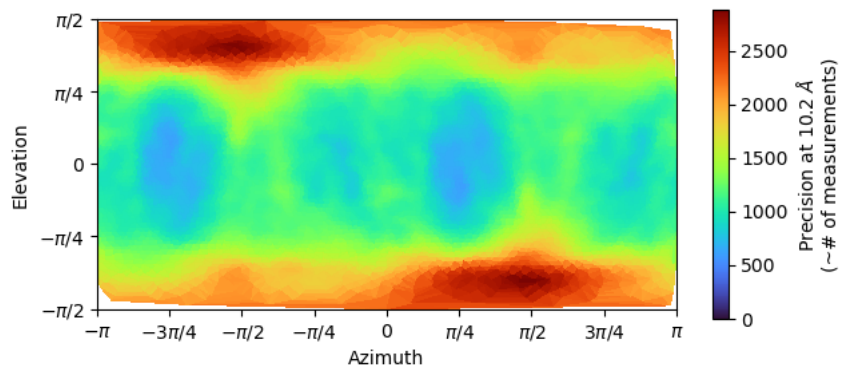
> [2026-06-09 15:40:24.79 GMT-4] Noise Model Iteration 001



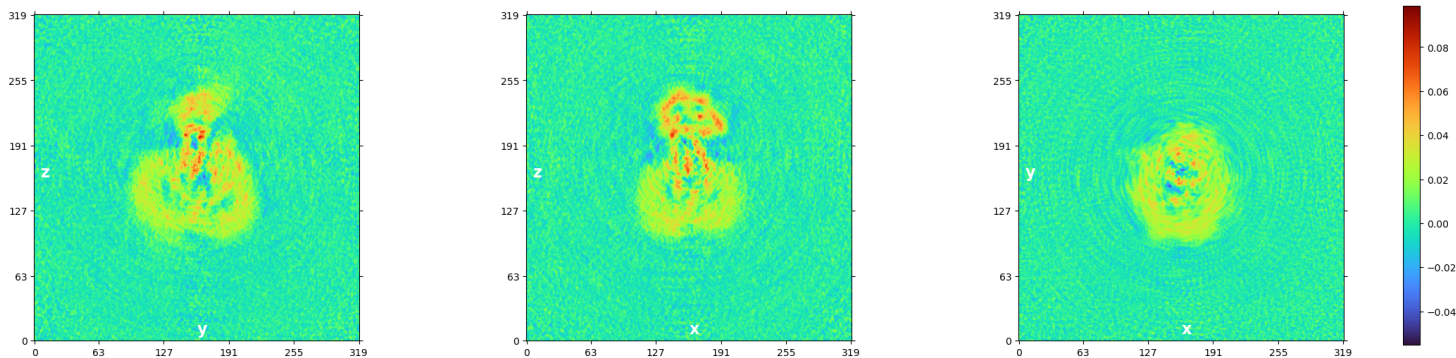
> [2026-06-09 15:40:25.58 GMT-4] Viewing Direction Distribution Iteration 001



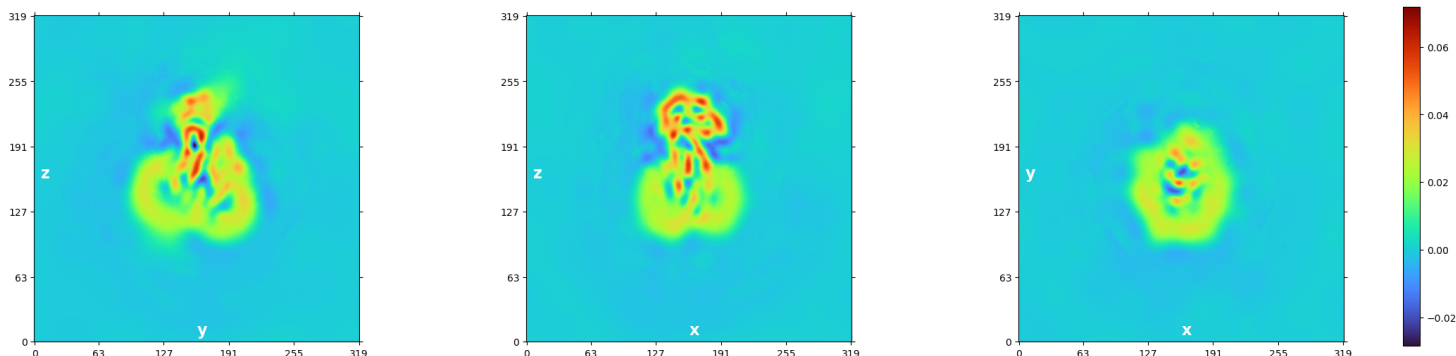
> [2026-06-09 15:40:27.16 GMT-4] Posterior Precision Directional Distribution Iteration 001



```
> [2026-06-09 15:40:27.16 GMT-4] [CPU: 10.24 GB] [Avail: 312.47 GB] Done in 18.988s.
> [2026-06-09 15:40:27.17 GMT-4] [CPU: 10.24 GB] [Avail: 312.47 GB] Outputting files..
> [2026-06-09 15:40:42.80 GMT-4] [CPU: 10.15 GB] [Avail: 313.98 GB] Done in 15.628s.
> [2026-06-09 15:40:42.80 GMT-4] [CPU: 10.15 GB] [Avail: 313.98 GB] Done iteration 1 in 160.792s. Total time so far 333.154s
> [2026-06-09 15:40:43.92 GMT-4] [CPU: 10.15 GB] [Avail: 313.95 GB] ----- Start Iteration 2
> [2026-06-09 15:40:43.92 GMT-4] [CPU: 10.15 GB] [Avail: 313.95 GB] Using Max Alignment Radius 33.580 (7.909A)
> [2026-06-09 15:40:43.92 GMT-4] [CPU: 10.15 GB] [Avail: 313.95 GB] Auto batchsize: 2690 in each split
> [2026-06-09 15:40:43.92 GMT-4] [CPU: 10.15 GB] [Avail: 313.95 GB] Using dynamic mask.
> [2026-06-09 15:41:14.42 GMT-4] [CPU: 10.15 GB] [Avail: 312.78 GB] -- THR 1 BATCH 500 NUM 845 TOTAL 3.9648489 ELAPSED 20.031979
--
> [2026-06-09 15:41:37.09 GMT-4] [CPU: 11.00 GB] [Avail: 313.07 GB] Processed 5294.000 images in 22.668s.
> [2026-06-09 15:41:40.52 GMT-4] [CPU: 11.00 GB] [Avail: 312.94 GB] Computing FSCs...
> [2026-06-09 15:41:40.53 GMT-4] [CPU: 11.00 GB] [Avail: 312.94 GB] Using full box size 320, downsampled box size 160, with low
memory mode disabled.
> [2026-06-09 15:41:40.53 GMT-4] [CPU: 11.00 GB] [Avail: 312.94 GB] Computing FFTs on GPU.
> [2026-06-09 15:41:42.77 GMT-4] [CPU: 11.00 GB] [Avail: 312.85 GB] Done in 2.243s
> [2026-06-09 15:41:42.77 GMT-4] [CPU: 11.00 GB] [Avail: 312.85 GB] Computing cFSCs...
> [2026-06-09 15:41:45.17 GMT-4] [CPU: 11.00 GB] [Avail: 312.76 GB] Done in 2.394s
> [2026-06-09 15:41:45.17 GMT-4] [CPU: 11.00 GB] [Avail: 312.76 GB] Using Filter Radius 36.198 (7.337A) | Previous: 33.580
(7.909A)
> [2026-06-09 15:41:51.22 GMT-4] [CPU: 10.25 GB] [Avail: 313.29 GB] Non-uniform regularization with compute option: GPU
> [2026-06-09 15:41:51.22 GMT-4] [CPU: 10.25 GB] [Avail: 313.29 GB] Running local cross validation for A ...
> [2026-06-09 15:42:12.69 GMT-4] [CPU: 10.50 GB] [Avail: 312.20 GB] Local cross validation A done in 21.466s
> [2026-06-09 15:42:14.37 GMT-4] FSC Filtered Side A
```



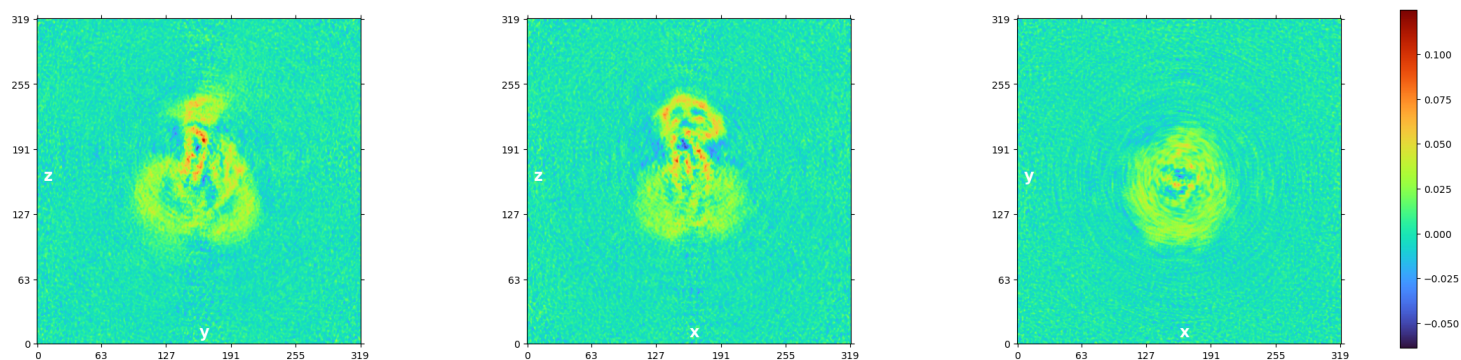
> [2026-06-09 15:42:15.55 GMT-4] CV Filtered Side A



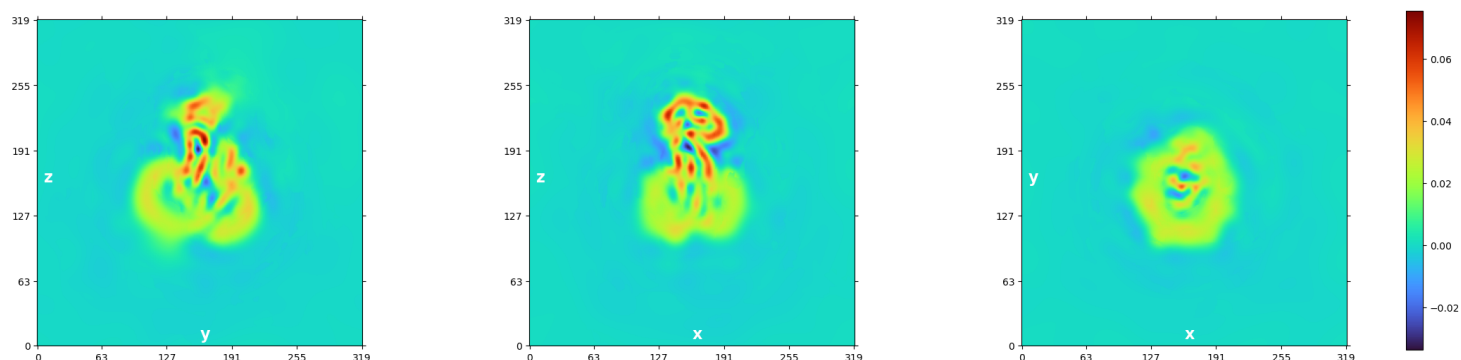
> [2026-06-09 15:42:15.56 GMT-4] [CPU: 10.50 GB] [Avail: 312.08 GB] Running local cross validation for B ...

> [2026-06-09 15:42:37.32 GMT-4] [CPU: 10.75 GB] [Avail: 312.99 GB] Local cross validation B done in 21.758s

> [2026-06-09 15:42:38.71 GMT-4] FSC Filtered Side B



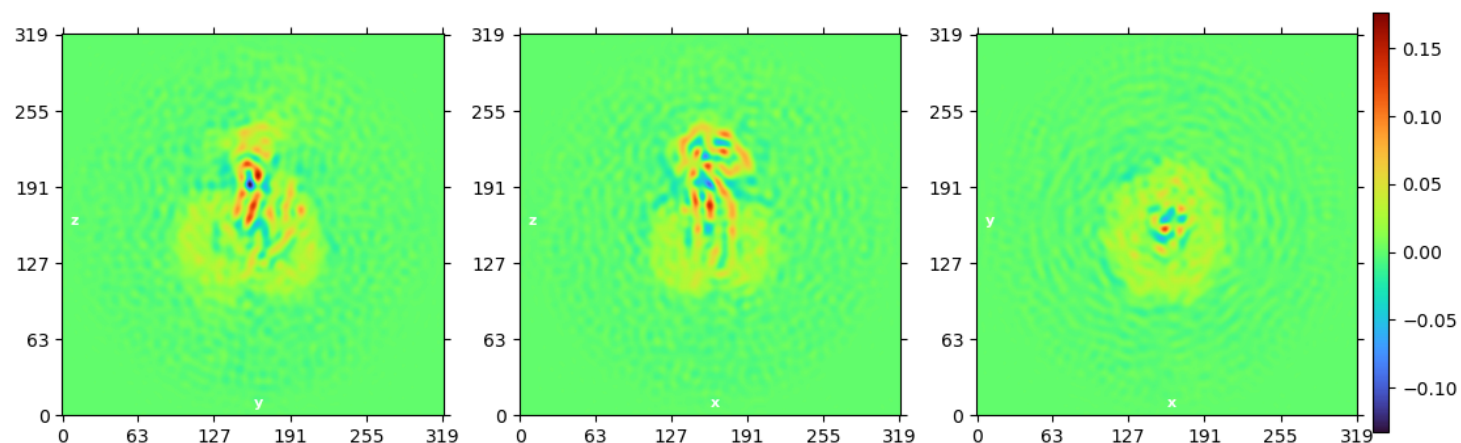
> [2026-06-09 15:42:39.65 GMT-4] CV Filtered Side B



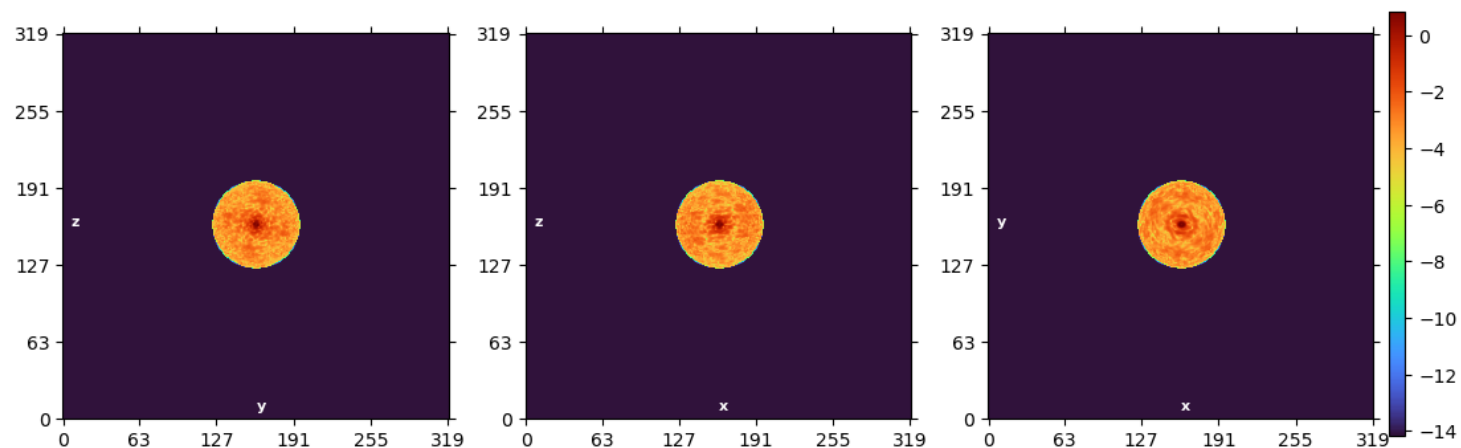
> [2026-06-09 15:42:49.67 GMT-4] [CPU: 10.72 GB] [Avail: 312.54 GB] Estimated Bfactor: -527.7

> [2026-06-09 15:42:49.68 GMT-4] [CPU: 10.72 GB] [Avail: 312.54 GB] Plotting..

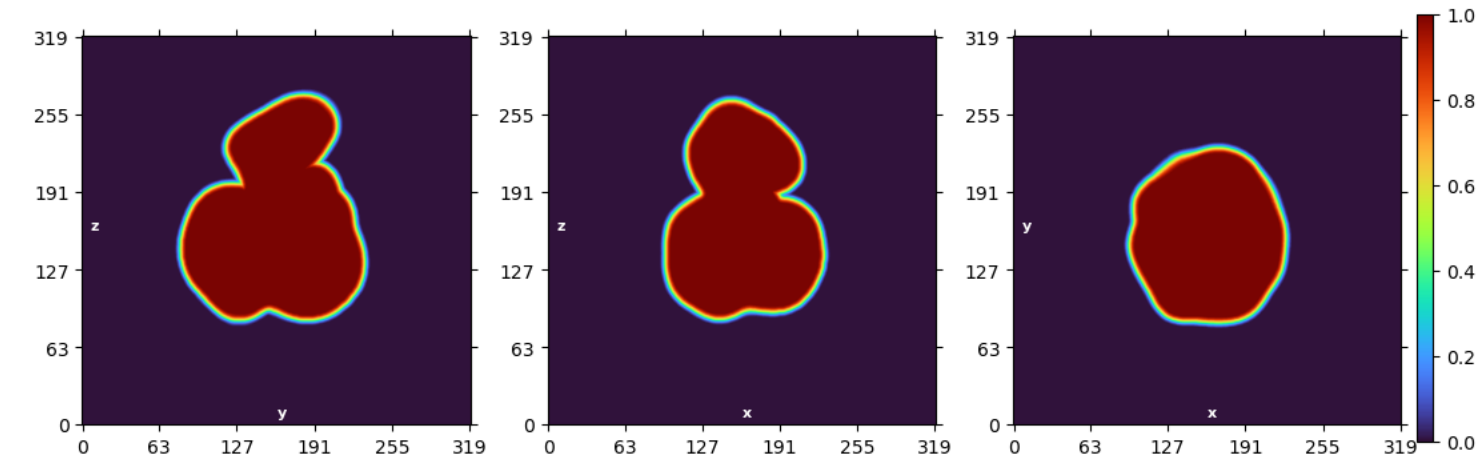
> [2026-06-09 15:42:55.29 GMT-4] Real Space Slices Iteration 002



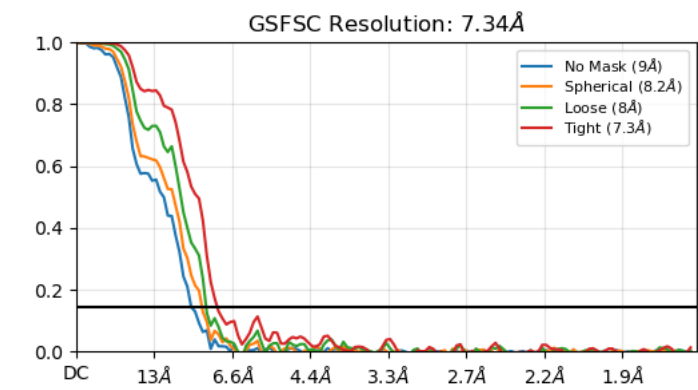
> [2026-06-09 15:42:56.40 GMT-4] Fourier Space Slices Iteration 002



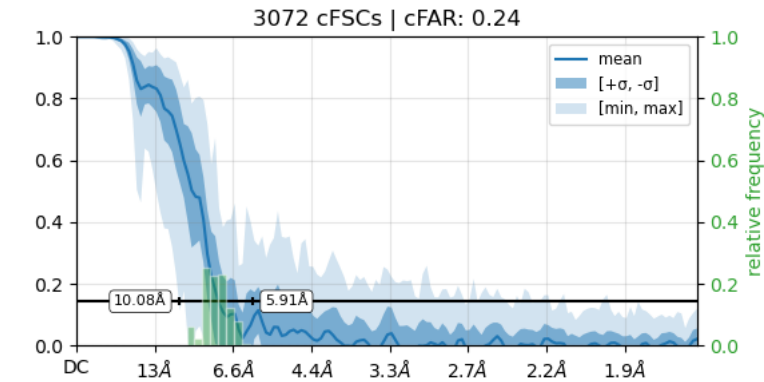
> [2026-06-09 15:42:58.16 GMT-4] Real Space Mask Slices Iteration 002



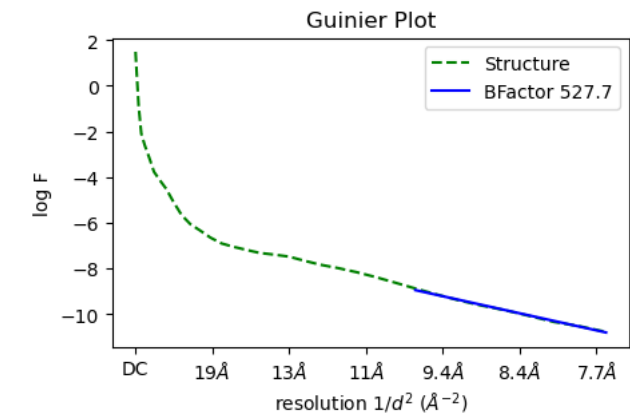
> [2026-06-09 15:42:58.58 GMT-4] FSC Iteration 002



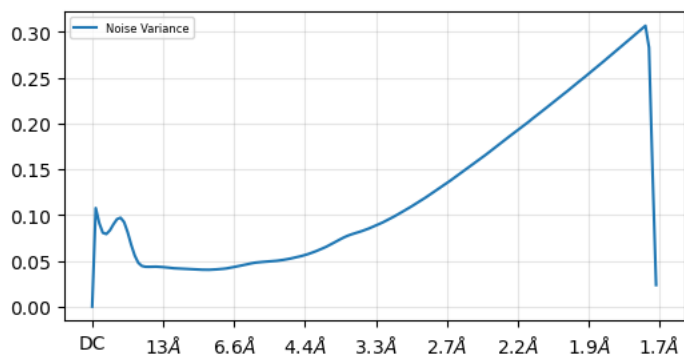
> [2026-06-09 15:42:59.20 GMT-4] cFSCs (Half-angle: 20°) Iteration 002, with tight mask



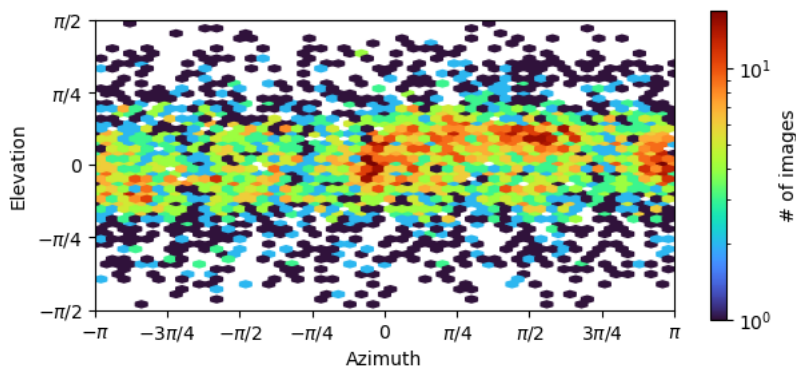
> [2026-06-09 15:43:04.49 GMT-4] Guinier Plot Iteration 002



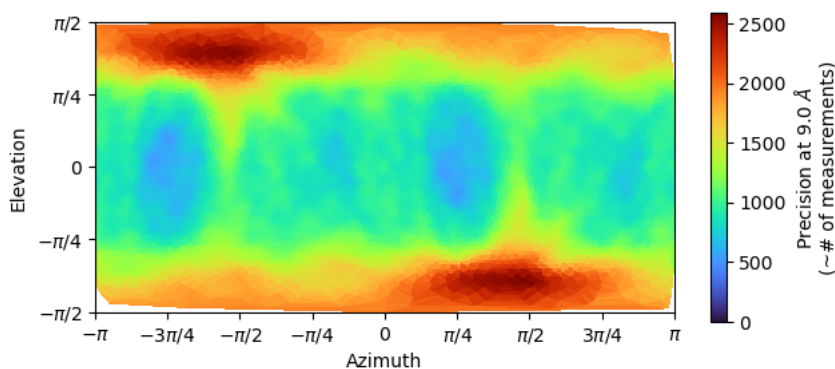
> [2026-06-09 15:43:04.79 GMT-4] Noise Model Iteration 002



> [2026-06-09 15:43:05.56 GMT-4] Viewing Direction Distribution Iteration 002



> [2026-06-09 15:43:06.83 GMT-4] Posterior Precision Directional Distribution Iteration 002



```
> [2026-06-09 15:43:06.83 GMT-4] [CPU: 10.75 GB] [Avail: 311.84 GB] Done in 17.151s.
> [2026-06-09 15:43:06.84 GMT-4] [CPU: 10.75 GB] [Avail: 311.84 GB] Outputting files..
> [2026-06-09 15:43:21.82 GMT-4] [CPU: 10.66 GB] [Avail: 313.35 GB] Done in 14.979s.
> [2026-06-09 15:43:21.82 GMT-4] [CPU: 10.66 GB] [Avail: 313.35 GB] Done iteration 2 in 157.902s. Total time so far 492.174s
> [2026-06-09 15:43:22.56 GMT-4] [CPU: 10.66 GB] [Avail: 313.32 GB] ----- Start Iteration 3
> [2026-06-09 15:43:22.57 GMT-4] [CPU: 10.66 GB] [Avail: 313.32 GB] Using Max Alignment Radius 36.198 (7.337A)
> [2026-06-09 15:43:22.57 GMT-4] [CPU: 10.66 GB] [Avail: 313.32 GB] Using Full Dataset (split 2604 in A, 2690 in B)
> [2026-06-09 15:43:22.57 GMT-4] [CPU: 10.66 GB] [Avail: 313.32 GB] Using dynamic mask.
> [2026-06-09 15:43:51.17 GMT-4] [CPU: 10.66 GB] [Avail: 312.21 GB] -- THR 0 BATCH 500 NUM 845 TOTAL 2.8674521 ELAPSED 18.340065
--
> [2026-06-09 15:44:11.52 GMT-4] [CPU: 11.41 GB] [Avail: 312.66 GB] Processed 5294.000 images in 20.346s.
> [2026-06-09 15:44:14.64 GMT-4] [CPU: 11.41 GB] [Avail: 312.55 GB] Computing FSCs...
> [2026-06-09 15:44:14.65 GMT-4] [CPU: 11.41 GB] [Avail: 312.55 GB] Using full box size 320, downsampled box size 160, with low
memory mode disabled.
> [2026-06-09 15:44:14.65 GMT-4] [CPU: 11.41 GB] [Avail: 312.55 GB] Computing FFTs on GPU.
> [2026-06-09 15:44:18.18 GMT-4] [CPU: 11.41 GB] [Avail: 312.40 GB] Done in 3.531s
> [2026-06-09 15:44:18.20 GMT-4] [CPU: 11.41 GB] [Avail: 312.41 GB] Optimizing FSC Mask...
> [2026-06-09 15:44:18.20 GMT-4] [CPU: 11.41 GB] [Avail: 312.41 GB] Using full box size 320, downsampled box size 160, with low
memory mode disabled.
> [2026-06-09 15:44:18.21 GMT-4] [CPU: 11.41 GB] [Avail: 312.41 GB] Computing FFTs on GPU.
> [2026-06-09 15:44:20.06 GMT-4] [CPU: 11.66 GB] [Avail: 312.08 GB] Using full box size 320, downsampled box size 160, with low
memory mode disabled.
> [2026-06-09 15:44:20.07 GMT-4] [CPU: 11.66 GB] [Avail: 312.08 GB] Computing FFTs on GPU.
> [2026-06-09 15:44:21.91 GMT-4] [CPU: 11.66 GB] [Avail: 312.00 GB] Using full box size 320, downsampled box size 160, with low
```



```

> [2026-06-09 15:44:21.91 GMT-4] [CPU: 11.66 GB] [Avail: 312.00 GB] Computing FFTs on GPU.
> [2026-06-09 15:44:23.87 GMT-4] [CPU: 11.66 GB] [Avail: 311.93 GB] Using full box size 320, downsampled box size 160, with low
memory mode disabled.

> [2026-06-09 15:44:23.87 GMT-4] [CPU: 11.66 GB] [Avail: 311.93 GB] Computing FFTs on GPU.
> [2026-06-09 15:44:25.80 GMT-4] [CPU: 11.66 GB] [Avail: 311.85 GB] Using full box size 320, downsampled box size 160, with low
memory mode disabled.

> [2026-06-09 15:44:25.80 GMT-4] [CPU: 11.66 GB] [Avail: 311.85 GB] Computing FFTs on GPU.
> [2026-06-09 15:44:27.65 GMT-4] [CPU: 11.66 GB] [Avail: 311.78 GB] Using full box size 320, downsampled box size 160, with low
memory mode disabled.

> [2026-06-09 15:44:27.65 GMT-4] [CPU: 11.66 GB] [Avail: 311.78 GB] Computing FFTs on GPU.
> [2026-06-09 15:44:29.62 GMT-4] [CPU: 11.66 GB] [Avail: 311.70 GB] Using full box size 320, downsampled box size 160, with low
memory mode disabled.

> [2026-06-09 15:44:29.62 GMT-4] [CPU: 11.66 GB] [Avail: 311.70 GB] Computing FFTs on GPU.
> [2026-06-09 15:44:31.48 GMT-4] [CPU: 11.66 GB] [Avail: 311.63 GB] Using full box size 320, downsampled box size 160, with low
memory mode disabled.

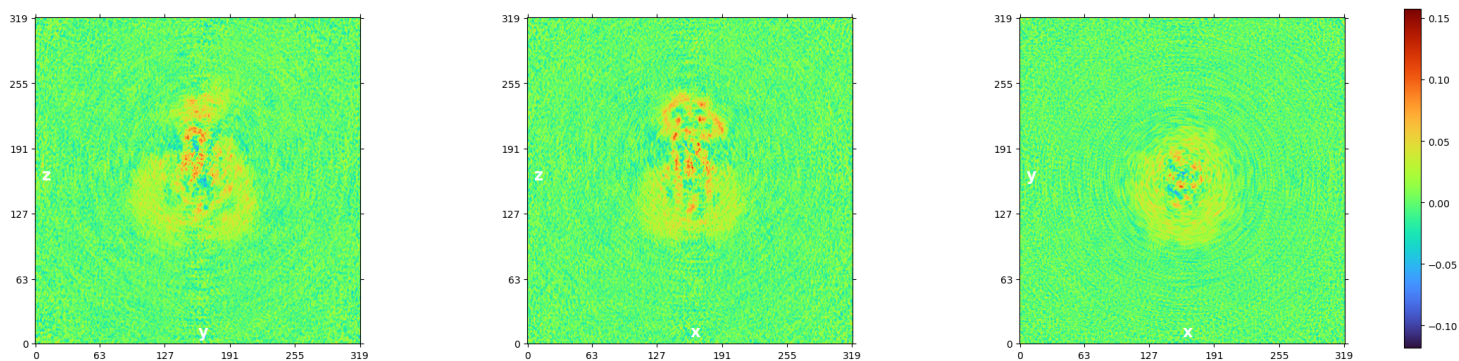
> [2026-06-09 15:44:31.48 GMT-4] [CPU: 11.66 GB] [Avail: 311.63 GB] Computing FFTs on GPU.
> [2026-06-09 15:44:33.42 GMT-4] [CPU: 11.66 GB] [Avail: 311.55 GB] Using full box size 320, downsampled box size 160, with low
memory mode disabled.

> [2026-06-09 15:44:33.42 GMT-4] [CPU: 11.66 GB] [Avail: 311.55 GB] Computing FFTs on GPU.
> [2026-06-09 15:44:35.39 GMT-4] [CPU: 11.66 GB] [Avail: 311.48 GB] Using full box size 320, downsampled box size 160, with low
memory mode disabled.

> [2026-06-09 15:44:35.39 GMT-4] [CPU: 11.66 GB] [Avail: 311.48 GB] Computing FFTs on GPU.
> [2026-06-09 15:44:37.27 GMT-4] [CPU: 11.66 GB] [Avail: 311.40 GB] Done in 19.066s
> [2026-06-09 15:44:37.27 GMT-4] [CPU: 11.66 GB] [Avail: 311.40 GB] Computing cFSCs...
> [2026-06-09 15:44:39.06 GMT-4] [CPU: 11.66 GB] [Avail: 311.34 GB] Done in 1.789s
> [2026-06-09 15:44:39.07 GMT-4] [CPU: 11.66 GB] [Avail: 311.34 GB] Using Filter Radius 38.107 (6.970A) | Previous: 36.198
(7.337A)

> [2026-06-09 15:44:45.17 GMT-4] [CPU: 10.91 GB] [Avail: 311.87 GB] Non-uniform regularization with compute option: GPU
> [2026-06-09 15:44:45.18 GMT-4] [CPU: 10.91 GB] [Avail: 311.87 GB] Running local cross validation for A ...
> [2026-06-09 15:45:07.88 GMT-4] [CPU: 11.16 GB] [Avail: 312.72 GB] Local cross validation A done in 22.696s
> [2026-06-09 15:45:09.25 GMT-4] FSC Filtered Side A

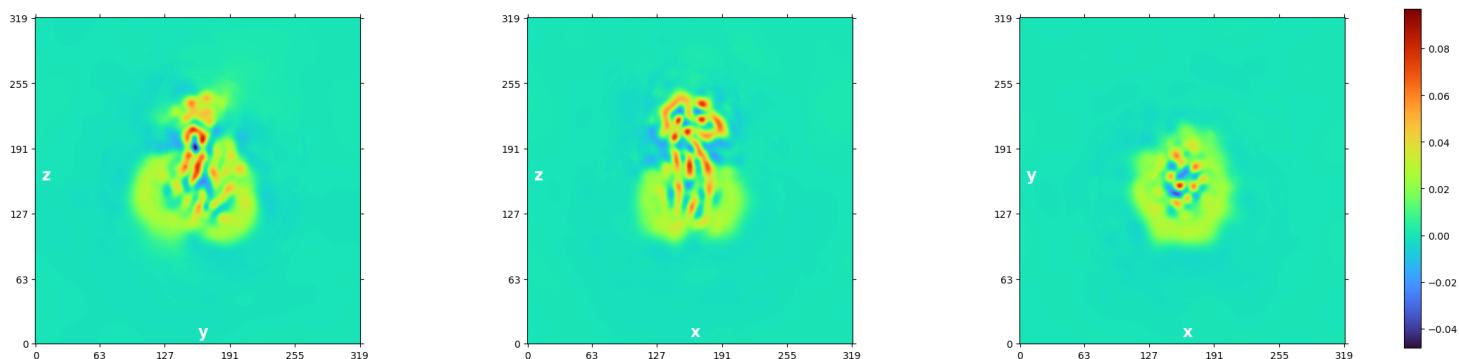
```



```

> [2026-06-09 15:45:10.45 GMT-4] CV Filtered Side A

```

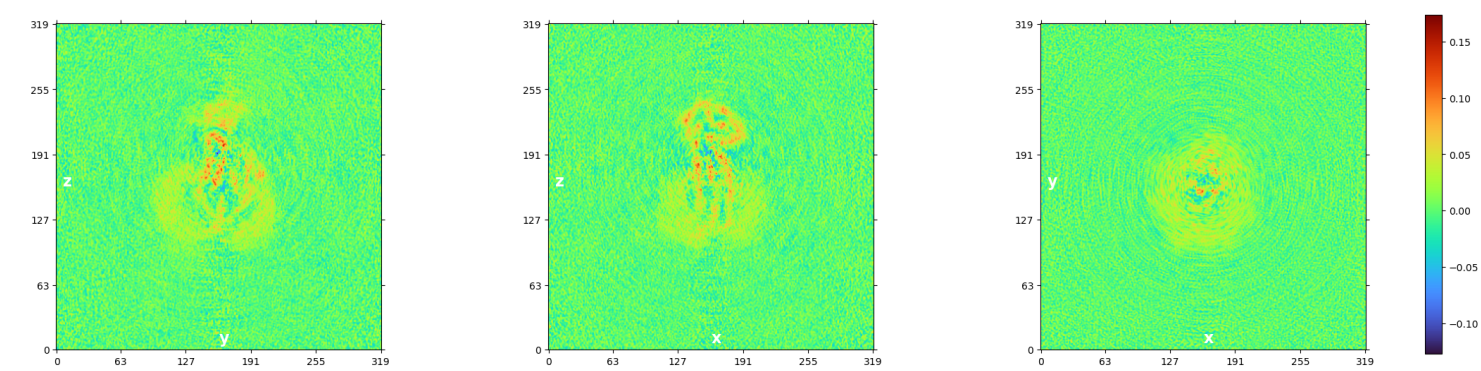


```

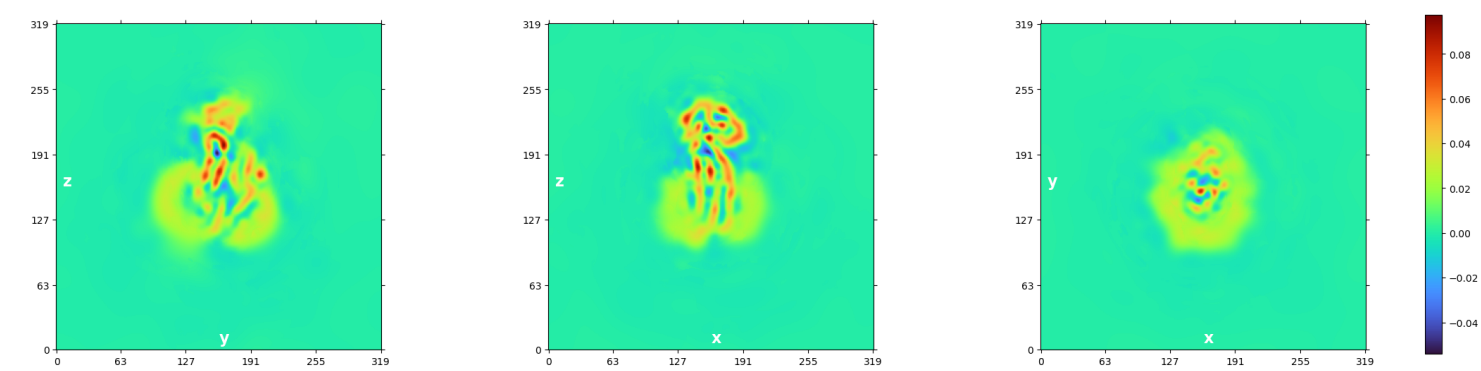
> [2026-06-09 15:45:10.46 GMT-4] [CPU: 11.16 GB] [Avail: 312.62 GB] Running local cross validation for B ...
> [2026-06-09 15:45:31.80 GMT-4] [CPU: 11.41 GB] [Avail: 311.54 GB] Local cross validation B done in 21.340s

```

> [2026-06-09 15:45:33.05 GMT-4] FSC Filtered Side B



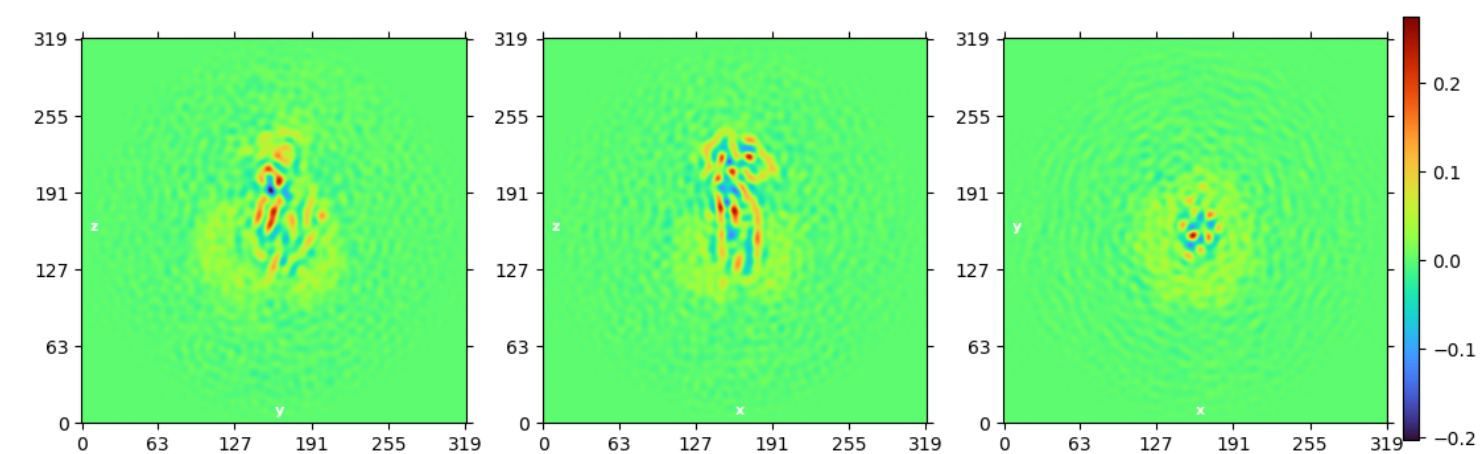
> [2026-06-09 15:45:34.22 GMT-4] CV Filtered Side B



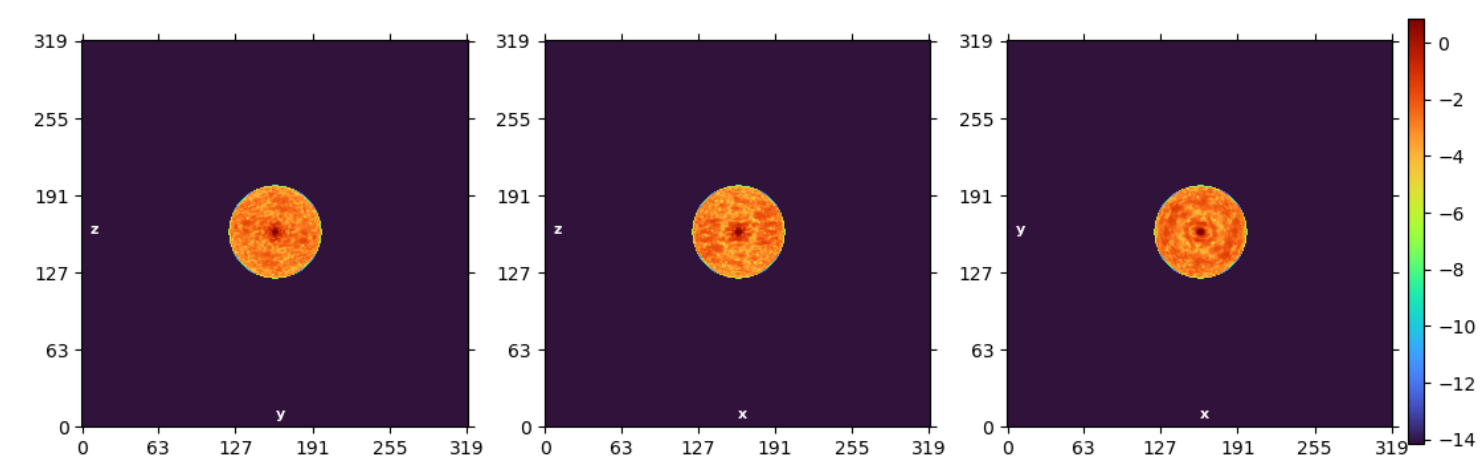
> [2026-06-09 15:45:44.27 GMT-4] [CPU: 11.38 GB] [Avail: 311.09 GB] Estimated Bfactor: -534.1

> [2026-06-09 15:45:44.28 GMT-4] [CPU: 11.38 GB] [Avail: 311.09 GB] Plotting..

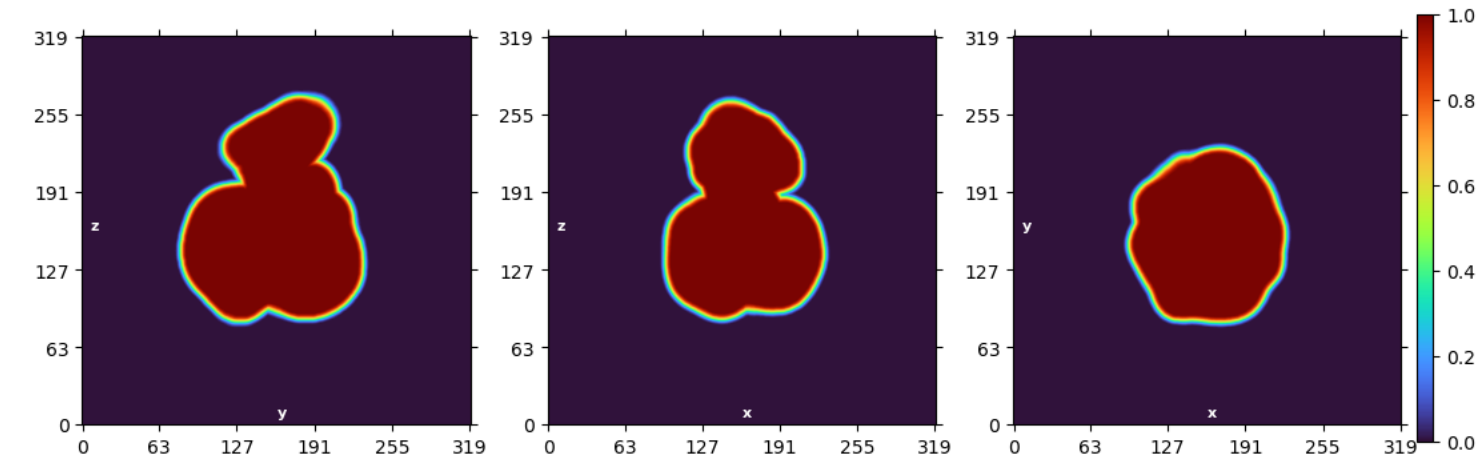
> [2026-06-09 15:45:50.51 GMT-4] Real Space Slices Iteration 003



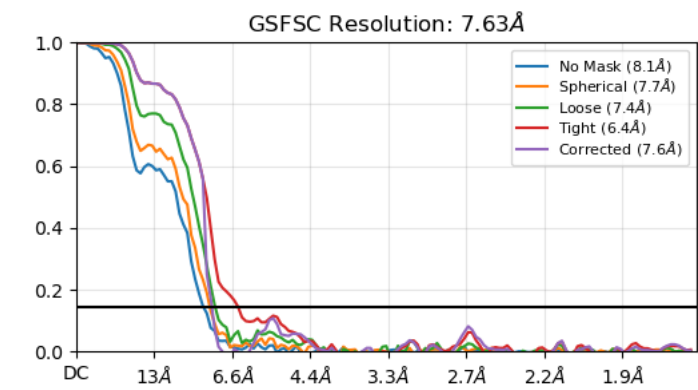
> [2026-06-09 15:45:51.63 GMT-4] Fourier Space Slices Iteration 003



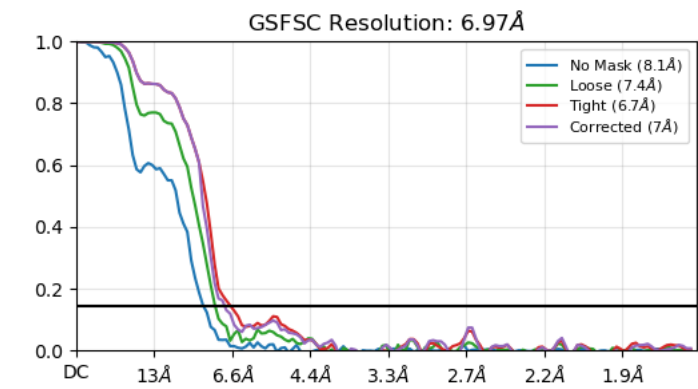
> [2026-06-09 15:45:53.46 GMT-4] Real Space Mask Slices Iteration 003



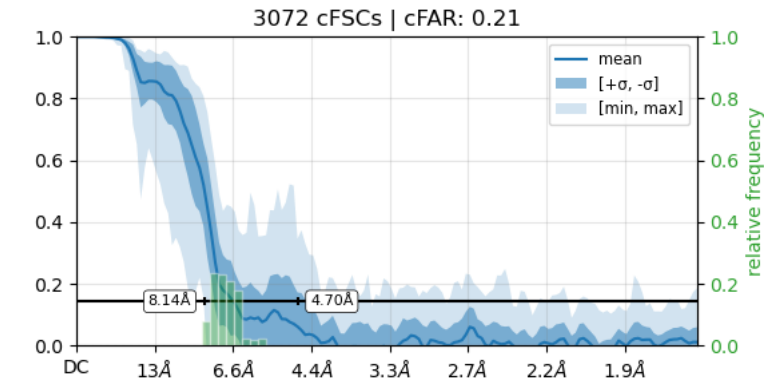
> [2026-06-09 15:45:53.90 GMT-4] FSC Iteration 003



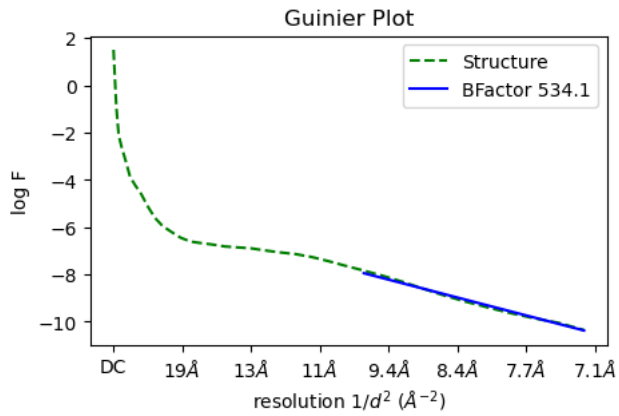
> [2026-06-09 15:45:54.32 GMT-4] FSC Iteration 003, after FSC-mask auto-tightening



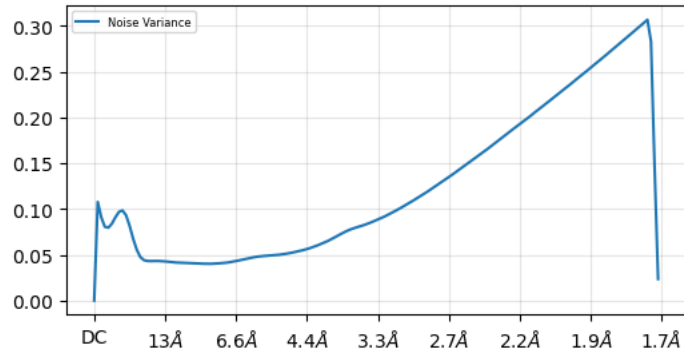
> [2026-06-09 15:45:54.93 GMT-4] cFSCs (Half-angle: 20°) Iteration 003, with autotight mask



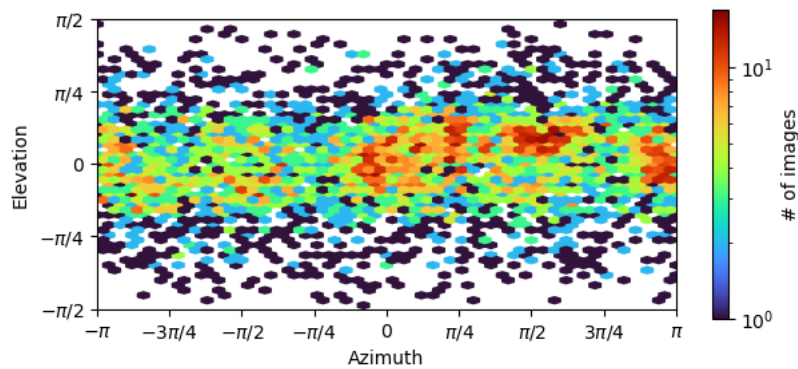
> [2026-06-09 15:46:00.23 GMT-4] Guinier Plot Iteration 003



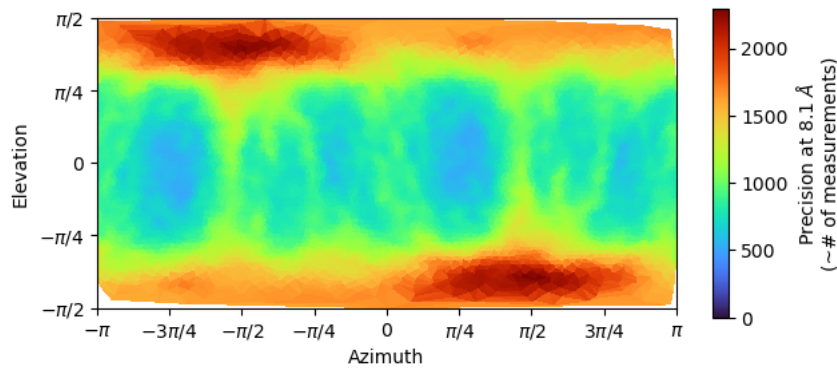
> [2026-06-09 15:46:00.54 GMT-4] Noise Model Iteration 003



> [2026-06-09 15:46:01.19 GMT-4] Viewing Direction Distribution Iteration 003



> [2026-06-09 15:46:02.56 GMT-4] Posterior Precision Directional Distribution Iteration 003



> [2026-06-09 15:46:02.56 GMT-4] [CPU: 11.41 GB] [Avail: 312.32 GB] Done in 18.280s.

> [2026-06-09 15:46:02.56 GMT-4] [CPU: 11.41 GB] [Avail: 312.32 GB] Outputting files..

> [2026-06-09 15:46:19.51 GMT-4] [CPU: 11.32 GB] [Avail: 311.76 GB] Done in 16.939s.

> [2026-06-09 15:46:19.51 GMT-4] [CPU: 11.32 GB] [Avail: 311.76 GB] Done iteration 3 in 176.940s. Total time so far 669.861s

> [2026-06-09 15:46:20.18 GMT-4] [CPU: 11.32 GB] [Avail: 311.73 GB] ===== Done Refinement =====

> [2026-06-09 15:46:20.19 GMT-4] [CPU: 11.32 GB] [Avail: 311.73 GB] Note that the output structure from refinement has been lowpass filtered to the gold-standard FSC resolution estimated above. However, the structure probably needs sharpening in order to best visualize high resolution features. This can be


```
done with the Sharpening task (as a new experiment).
> [2026-06-09 15:46:20.19 GMT-4] [CPU: 11.32 GB] [Avail: 311.73 GB] Full run took 670.547s
> [2026-06-09 15:46:20.83 GMT-4] [CPU: 5.46 GB] [Avail: 317.57 GB] -----
-
> [2026-06-09 15:46:20.84 GMT-4] [CPU: 5.46 GB] [Avail: 317.57 GB] Compiling job outputs...
> [2026-06-09 15:46:20.84 GMT-4] [CPU: 5.46 GB] [Avail: 317.57 GB] Passing through outputs for output group particles from input
group particles
> [2026-06-09 15:46:20.89 GMT-4] [CPU: 5.46 GB] [Avail: 317.57 GB] This job outputted results ['alignments3D', 'ctf']
> [2026-06-09 15:46:20.89 GMT-4] [CPU: 5.46 GB] [Avail: 317.57 GB]   Loaded output dset with 5294 items
> [2026-06-09 15:46:20.89 GMT-4] [CPU: 5.46 GB] [Avail: 317.57 GB] Passthrough results ['blob', 'alignments3D_multi']
> [2026-06-09 15:46:20.95 GMT-4] [CPU: 5.46 GB] [Avail: 317.57 GB]   Loaded passthrough dset with 5294 items
> [2026-06-09 15:46:20.98 GMT-4] [CPU: 5.46 GB] [Avail: 317.57 GB]   Intersection of output and passthrough has 5294 items
> [2026-06-09 15:46:21.00 GMT-4] [CPU: 5.46 GB] [Avail: 317.57 GB]   Output dataset contains: ['alignments3D_multi', 'blob']
> [2026-06-09 15:46:21.01 GMT-4] [CPU: 5.46 GB] [Avail: 317.57 GB]   Outputting passthrough result blob
> [2026-06-09 15:46:21.02 GMT-4] [CPU: 5.46 GB] [Avail: 317.57 GB]   Outputting passthrough result alignments3D_multi
> [2026-06-09 15:46:21.03 GMT-4] [CPU: 5.46 GB] [Avail: 317.57 GB] Checking outputs for output group particles
> [2026-06-09 15:46:21.14 GMT-4] [CPU: 5.46 GB] [Avail: 317.56 GB] Updating job size...
> [2026-06-09 15:46:21.36 GMT-4] [CPU: 5.46 GB] [Avail: 317.55 GB] Exporting job and creating csg files...
> [2026-06-09 15:46:21.91 GMT-4] [CPU: 5.46 GB] [Avail: 317.53 GB] *****
**
> [2026-06-09 15:46:21.92 GMT-4] [CPU: 5.46 GB] [Avail: 317.53 GB] Job complete. Total time 672.30s
```